

Energy

ENERGY UNITS

Energy requirements for maintenance and milk production are expressed in net energy for lactation (NEL) units. The NEL system has been used by the National Research Council (NRC) for dairy cattle nutrient requirements for several editions because it uses a single energy unit (NEL) for both maintenance and milk production. The classical energy flow system used in animal nutrition for decades is as follows: gross energy (GE), digestible energy (DE), metabolizable energy (ME), and finally net energy. In the current version, the DE, ME, and NEL values of feeds are all considered as the actual amount of energy that would be provided based on the animal and diet; in other words, DE and ME are not the potentially maximum digested or metabolizable energy but the DE and ME expected in a given situation (cow and diet). Thus, the model is for evaluation and should be used with caution for formulation.

Based on work from the USDA Energy Metabolism Unit at Beltsville 50 years ago, the efficiency of using ME for maintenance (0.62) and milk production (0.64) was considered essentially the same (Flatt et al., 1965; Tyrrell and Moe, 1972). Using only the last two decades of work at Beltsville, as reported in Moraes et al. (2015), the conversion of ME to milk is 0.66, and the conversion of ME to body reserves to milk is similar. Therefore, one feed energy value (NEL) is used to express the requirements for maintenance, pregnancy, milk production, frame gain (growth), and changes in body reserves (tissue that is lost and gained during times of nutrient excess or deficiency) of adult cows. The efficiencies of using NEL for pregnancy, frame gain, and changes in body reserves are adjusted to fit within this system for adult cows, as discussed later. The energy requirements for cattle before their first parturition are given on an ME basis (see Chapters 10 and 11). As discussed in the seventh revised edition, one nutrient can alter the digestibility of other nutrients, and the conversion of DE to ME is altered by the composition

of the diet; therefore, ME and NEL values are not accurate or valid for individual feeds and should only be calculated for total diets.

ENERGY VALUES OF FEEDS

The method used to estimate feed and dietary energy values in this edition is similar to that used by NRC (2001) but includes significant modifications. The seventh edition did not give fixed NEL values for a feed; rather, NEL values of diets were based on the composition of feeds and diets and level of intake. In this edition, modifications were made to improve accuracy and account for more sources of variation than in the seventh edition. Deficiencies with the seventh revised edition that were addressed include the following:

- The digestibility discount as intake increased was too great (Huhtanen et al., 2009; White et al., 2017; de Souza et al., 2018). In NRC (2001), the digestibility discount was larger in diets that had higher basal total digestible nutrient (TDN) concentrations; TDN was essentially a proxy for dietary starch content, and diets with more starch often are consumed at greater intakes. The structure of this equation resulted in exaggerated negative effects on digestibility as starch content and intake were both increased.
- The digestibility discount was applied to the entire diet; however, intake does not affect digestibility of all nutrient fractions similarly.
- Protein was appropriately given a higher DE value than starch (5.65 compared to 4.2 Mcal/kg), but the ME value of excess protein was not correct. When protein is used as a fuel source, its ME value is similar to starch. The previous version overestimated the energy value of protein when fed in excess of requirement.
- The equations used to convert DE at production intakes to ME and ME to NEL had negative intercepts, which

likely resulted in underestimating the conversion efficiency of lower-energy diets.

- Level of intake was calculated as a multiple of maintenance energy, which results in a circular argument. Intake of NEL altered the efficiency of converting DE at maintenance intake to NEL, which in turn altered NEL intake.

Other improvements made from the seventh revised edition include the following:

- Nonfiber carbohydrate and TDN are no longer used. Many of the factors that affect starch digestibility have been quantified; therefore, including starch in the equation and adjusting for those factors should improve accuracy. The negative associative effects of starch on fiber digestibility have been quantified, and these are used in place of TDN to estimate digestibility discounts and DE.
- The base for DE calculations was set as a cow consuming dry matter (DM) at 3.5 percent of body weight (BW) and fed a diet with 26 percent starch. These values are the averages in the data set used to generate digestibility values. In the previous edition, the base was a cow fed at maintenance (approximately 1.2 percent of BW), which required substantial extrapolation of digestibility values.
- Rather than using an essentially constant efficiency for converting DE to ME, energy lost via urine and methane is now calculated using diet and animal characteristics, resulting in more variable efficiencies, which should increase accuracy over a wider array of diets.
- The energy values for protein are calculated using values derived from the protein system (i.e., rumen degradable protein [RDP], rumen undegradable protein [RUP], and digestibility of RUP). In the previous version, the energy value of protein was calculated independently of the protein system.

OVERALL ENERGY SCHEME

The overall approach (see Figure 3-1) used to estimate diet energy values is as follows: (1) feed is separated into fractions that mostly approximate uniform fractions, (2) gross energy values are calculated based on these fractions, (3) base digestibilities for each feed fraction are calculated assuming dry matter intake (DMI) at 3.5 percent of BW and a dietary starch content at 26 percent, (4) adjustments are made to base digestibility values for level of intake on a DMI/BW basis and for dietary starch, and (5) estimates of urinary energy (UE) and gas energy output are calculated based on diet and animal characteristics. Feed NEL values are no longer provided, even in the tables, as diet NEL supply must be based on the whole diet, and thus NEL values for individual feeds are misleading. The same is true for ME and, to a lesser extent, DE. Base DE values (i.e., DMI = 3.5 percent of BW and diet contains 26 percent starch) for feeds are in

Table 19-1; however, the DE of a diet formulated using table values likely will differ from one formulated using the model even if the composition of the ingredients is the same.

FEED FRACTIONS

The summative approach to estimating DE in the seventh revised edition was retained, but feed was separated into more fractions: neutral detergent fiber (NDF), starch, fatty acids (FAs), crude protein (CP) ($N \times 6.25$), ash, and residual organic matter (ROM). The FA fraction includes FAs with more than four carbons and specifically does not include the short-chain volatile FAs or lactic acid. The ROM fraction is DM not accounted for in the main feed fractions (Equation 3-1). This by-difference fraction contains water-soluble carbohydrates, ingested fermentation and other short-chain FA (e.g., acetic, butyric, and lactic acids), glycerol (both free and the glycerol moiety of triglycerides [TGs]), soluble fiber (pectins and gums), and any components not accounted for in the main feed fractions (e.g., tannins and waxes). The FA content of TGs includes an extra water molecule for the hydrolysis of each FA ester bond. Thus, the total mass of hydrolyzed FA and glycerol is 106 percent of the original TG mass for typical feed lipids. To estimate the amount of ROM (glycerol) in a TG, the mass of FA must be divided by 1.06. This correction may not account properly for the ROM content of some lipids, such as TGs with shorter-chain FA, phospholipids, and glycolipids, but these fractions are not generally measured, and any error would be small. In addition, the correction is not appropriate for FA from nonesterified sources as shown in Equation 3-1. The CP equivalent from supplemental nonprotein nitrogen (sNPNCP) is separated from CP when estimating energy values. The concentration of ROM is also adjusted (i.e., $181 / 281 = 0.64$) to correctly account for the mass of supplemental nonprotein nitrogen (NPN) (Equation 3-1). Without this correction, urea (281 percent CP and -181 percent ROM) would have a GE value of 8.6 kcal/g instead of 2.5 kcal/g. Detailed information regarding assays for these fractions is given in Chapter 18.

$$\text{ROM} = 100 - \text{Ash} - \text{NDF} - \text{Starch} \\ - (\text{FA} / \text{FatFactor}) - (\text{CP} - 0.64 \times \text{sNPNCP})$$

(Equation 3-1)

where FatFactor = 1 if Feedtype = fatty acid or FA soap and 1.06 for all other feeds, and values are a percentage of DM.

The NDF fraction is a heterogeneous mixture of carbohydrates, lignin, nitrogen-containing compounds, and ash. Because both ash and CP are included as unique fractions in the ROM equation, any ash and CP contained in the NDF fraction will be counted twice and will result in an under-estimation of the ROM fraction with an equal overestimation of the NDF fraction. For most feeds, CP comprises

<10 percent of NDF (but can approach 20 percent in some feeds), and ash comprises <2 percent of NDF but can be >7 percent in some feeds (Crocker et al., 1998). For mixed

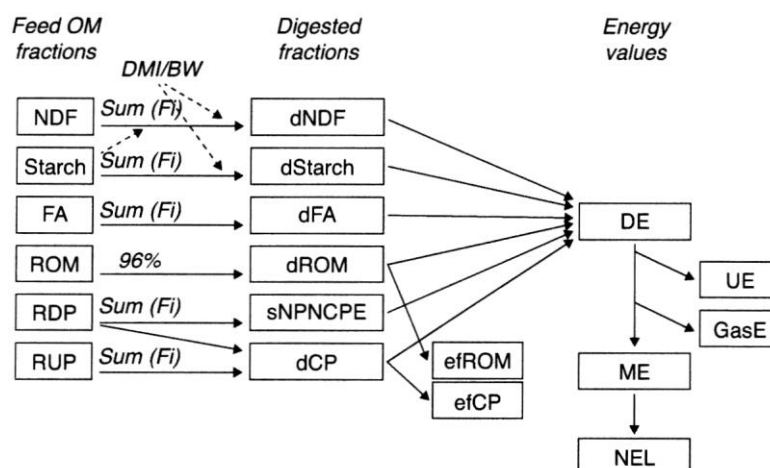


FIGURE 3-1 Feed energy supply system for dairy cattle. Each feed (Fi) is fractioned into NDF, starch, FAs (>4 carbons), ROM (mostly sugars, pectins, gums, the glycerol moiety of TGs, and fermentation acids), RDP, and RUP. These are converted to digested nutrients (dNDF, dStarch, dFA, dROM, and dCP) using base digestion coefficients for each individual feed. Starch and NDF digestion are adjusted for level of intake, and dNDF is further adjusted for the content of starch in the whole diet. The RDP that is from supplemental NPN on a CP-equivalent basis (sNPNCPE) is separated from dCP because it has a lower digestible energy (DE) value than dCP. The digested fractions are then converted to an apparent DE value by multiplying each digested fraction by its enthalpy of combustion and then subtracting endogenous fecal energy, which includes undigested microbial protein and endogenous fecal ROM (efCP, efROM). The amount of UE is estimated based on urinary nitrogen (UN), with UN a function of apparently digested CP and N captured. Gaseous energy (GasE) is a function of DMI and the content of FA and digested NDF in the diet. ME is calculated as DE minus UE, and GasE and is converted to NEL using a constant efficiency.

diets, the sum of ash and neutral detergent insoluble CP usually comprises 6 to 7 percent of the NDF (Tebbe et al., 2017), so that the ash and CP-free NDF in a diet average about 94 percent of the NDF value.

Although the double subtraction of neutral detergent insoluble ash and CP is incorrect, NDF, rather than ash- and CP-free NDF, is used in energy supply equations because:

1. Data on the concentrations of ash- and CP-free NDF for many feeds are limited.
2. Most publications that reported *in vivo* digestibility values for NDF did not subtract CP or ash from NDF; therefore, estimated NDF digestion coefficients of the current model can be compared directly to *in vivo* data.
3. Neutral detergent-insoluble CP and ash are usually quantitatively small fractions, and analytical precision is likely less for ash- and CP-free NDF than for NDF.
4. The true ROM digestibility and endogenous fecal ROM estimates are more precise when NDF is used rather than ash- and CP-free NDF (Tebbe et al., 2017).
5. The CP and ash corrections are not quantitatively important for estimating energy values for most diets because the errors largely cancel each other out. In Tebbe et al. (2017), diets with the greatest difference between ash- and protein-free NDF and NDF resulted in a difference of < 0.1 percent in the sum of digested NDF and digested ROM.

Gross Energy Values of Feed Fractions

Estimated GE of a feed or diet is calculated by multiplying the proportion of each fraction by its respective GE value and summing (Equation 3-2). This GE value serves as a useful reference for animal experiments.

Starch: 4.23 Mcal/kg

FA: 9.4 Mcal/kg

NDF: 4.2 Mcal/kg

ROM: 4.0 Mcal/kg (committee estimate assuming this fraction is predominantly sugars, organic acids [mostly lactic and acetic], glycerol, and soluble fiber)

CP (excluding supplemental NPN): 5.65 Mcal/kg

sNPNCPE: 0.89 Mcal/kg (calculated from the heat of combustion of urea at 2.5 Mcal/kg or 0.89 Mcal/kg CP equivalent)

The GE of a diet or feed (Mcal/kg) is calculated as

$$\begin{aligned} \text{GE}_{\text{DM of feed}} = & 0.042 \times \text{NDF}_{\text{DM}} + 0.0423 \\ & \times \text{Starch}_{\text{DM}} + 0.040 \times \text{ROM}_{\text{DM}} \\ & + 0.094 \times \text{FA}_{\text{DM}} + 0.0565 \times (\text{CP}_{\text{DM}} \\ & - \text{sNPNCPE}_{\text{DM}}) + 0.0089 \times \text{sNPNCPE}_{\text{DM}} \end{aligned} \quad (\text{Equation 3-2})$$

where feed fractions are expressed as a percentage of DM.

ESTIMATING THE DIGESTIBLE ENERGY VALUE FOR FEEDS AND DIETS

True Digestibility Coefficients of Feed Fractions for Base Conditions

In the previous edition (NRC, 2001), DE of diets was estimated for a cow fed at maintenance and then discounted as DMI increased and as the energy concentration of the diet (expressed as TDN) increased. In this version, the base condition is for an animal with a DMI of 3.5 percent of BW and fed a diet with 26 percent starch. All diets are assumed to have adequate RDP to meet microbial requirements and adequate forage NDF to promote proper rumen conditions. The DMI and starch concentration for base conditions reflect the mean of the database. Inadequate data are available to accurately estimate effects of RDP or forage on digestibility. Under practical conditions, RDP is usually adequate, but if diets do not supply adequate RDP (see Chapter 6), diet energy values may be overestimated. Diets with inadequate forage NDF (see Chapter 5) can cause ruminal acidosis, resulting in lower than estimated NDF and energy digestibilities. Some of the negative effects of inadequate forage NDF on digestibility should be accounted for by the starch adjustment. Although most of the digestibility data used to develop equations are from Holstein cows, digestibility is usually not different between Holstein and Jersey cows (Aikman et al., 2008; Knowlton et al., 2010; Uddin et al., 2020).

Neutral Detergent Fiber

Because there is no endogenous fecal NDF, apparent and true digestibility of NDF are the same. NDF can be expressed as NDF, NDF on a CP-free basis, on an ash-free basis, or on a CP- and ash-free basis. Based on limited data (Tebbe et al., 2017), the digestibility of CP- and ash-free NDF is slightly greater than the digestibility of NDF, but the NDF concentration in the feed is less, so concentrations of digested NDF and digested CP- and ash-free NDF are similar. Thus, digested NDF is used to estimate energy. In the model, two methods can be used to estimate base digestibility of NDF; one is the lignin-based equation (Equation 3-3a) from the seventh revised edition, and the other uses 48-hour in vitro NDF digestibility (IVNDFD; Lopes et al., 2015, Equation 3-3b). Incubations for 48 hours were more accurate at estimating in vivo NDF digestibility by lactating cows fed ad libitum (ca. 24 kg DMI/d) than were 30-hour incubations (Lopes et al., 2015). Inadequate IVNDFD data from published studies that measured in vivo digestibility were available to make a robust comparison of the two methods. The evaluations in Chapter 20 are based on the lignin equation, and the lignin method is the default method used in the model; however, the user has the option of using the IVNDFD equation or directly entering an NDF digestibility value.

$$\begin{aligned} \text{Digested proportion of NDF at base (dNDF_NDF_base)} \\ = \{0.75 \times (\text{NDF_DM} - \text{Lignin_DM}) \\ \times [1 - (\text{Lignin_DM} / \text{NDF_DM})^{0.667}]\} / \text{NDF_DM} \end{aligned}$$

(Equation 3-3a)

where nutrients are expressed as a percentage of DM.

$$\begin{aligned} \text{Digested proportion of NDF at base} \\ (\text{dNDF_NDF_base}) = 0.12 + 0.61 \times \text{IVNDFD} \end{aligned}$$

(Equation 3-3b)

where IVNDFD is 48-hour in vitro digestion expressed as a proportion of NDF.

For the common macronutrients, the in vivo digestibility of NDF is the most variable, and more research is needed to improve its estimation using commercially applicable laboratory methods. This research must include comparisons of laboratory-based estimates to in vivo measurements in dairy cows fed typical diets (e.g., Kendall et al., 2009).

Starch

Starch digestibility is dependent on innate properties of starch granules in grains, on the timing of harvest, and on mechanical processing that occurs postharvest (see Chapter 5). Starch digestibilities of the major starch sources are shown in Table 3-1, and these values are used as base starch digestibilities in the electronic feed library.

Protein

The digestibility of protein is based on the protein model, so that the true total-tract digestibility of the protein in a feed is the sum of RDP and the digested portion of RUP (dRUP). The proportion of protein degraded in the rumen and proportion of RUP that is digested are not dependent on DMI (see Chapter 6); therefore, protein digestibility in the model is not affected by intake. The committee recognizes the possible error in this assumption. However, given that most potentially degraded protein that is undegraded because of a high passage rate will probably be digested in the total tract, this error is likely small when calculating DE. A recent meta-analysis by White et al. (2017) supports the idea that the total-tract digestibility of protein is less affected by intake than that of other nutrients.

$$\begin{aligned} \text{Proportion of digested CP (dCP_CP)} = (\text{RDP_DM} \\ + \text{dRUP_DM}) / \text{CP_DM} \end{aligned}$$

(Equation 3-4)

where RDP, dRUP, and CP are a percentage of DM.

Residual Organic Matter

Based on the Lucas test, ROM is a uniform feed fraction with a high true digestibility and an endogenous fecal frac-

TABLE 3-1 Proportion of Starch Digested at Base (dStarch_Starch_base) for Various Starch Sources^a

Feed	dStarch_Starch_base
Default	0.91
Com grain, dry, fine grind (<1,250 μm) ^b	0.92
Com grain, dry, medium grind (1,500 to 3,250 μm)	0.89
Com grain, dry, coarse grind (>3,500 μm)	0.77
Com grain, high moisture, fine grind (<2,000 μm , mean = 1,450 μm)	0.96
Com grain, high moisture, coarse grind (>2,500 μm , mean = 3,630 μm)	0.90
Com grain, steam flaked	0.94
Sorghum grain, dry, ground	0.83
Sorghum grain, steam flaked	0.94
Com silage <30 percent DM	0.91
Com silage 32-37 percent DM	0.89
Com silage >40 percent DM	0.85
Grain sorghum silage ^c	0.85
Barley, steam rolled	0.94
Barley, ground	0.91
Wheat	0.93

^a Coefficients were derived from experiments, reviews, and meta-analyses using lactating dairy cows (Bal et al., 1997; Cammell et al., 2000a,b; Firkins et al., 2001; Ferraretto et al., 2013).

^b Because of incomplete data, particle size classifications for com grain are not continuous. For com with particle sizes not listed, interpolation can be used.

^c Based on data from beef cattle (Gutierrez et al., 1982; Hart, 1987).

tion whether calculated using the standard NDF value or ash- and CP-free NDF (Tebbe et al., 2017). The true digestibility of ROM calculated with NDF was 0.96 and was set as the base digestibility (dROM_ROM_base) in the model.

Fatty Acids

The base digestibility of FAs (dFA_FA_base) is set at 0.73 for most feeds; however, for supplemental fat sources, true digestibility is dependent on the source of FAs and was based on published digestion data (see Table 4-1; Chapter 4). Digestibility for FAs is affected by FA saturation and length and perhaps by interactions among different FAs and can be depressed by high-fat diets. Because the FA profile of total diets will vary less than the FA profile of feeds, the true digestibility of FAs among basal diets is also likely less variable. Therefore, the committee decided to assign all basal feeds (excluding fat supplements) the same true digestibility for FAs, which represents the average true FA digestibility of mixed diets. Digestibility of FAs from different fat sources and supplements is discussed in Chapter 4.

Adjustments to the Base Digestibilities for Intake and Diet Composition

The digestibilities of NDF and starch are adjusted for level of intake, and NDF digestibility is also adjusted for starch

content of the total diet. The depression in digestibility as intake increases has long been recognized (Tyrrell and Moe, 1975) and was included either implicitly (NRC, 1989) or explicitly in previous editions (NRC, 2001). In NRC (2001), the intake discount was greater with higher baseline digestibilities. The overall depression in DM digestibility with increasing intake was overestimated in NRC (2001) (Huhtanen et al., 2009; White et al., 2017; de Souza et al., 2018). One likely reason for this overestimation was that level of intake and baseline digestibilities are confounded.

Neutral Detergent Fiber

The committee considered several approaches but adopted the adjustments to NDF digestibility from a meta-analysis of individual cow data from multiple studies at multiple locations conducted by de Souza et al. (2018) with modifications. In de Souza et al. (2018), digestibility of NDF in response to intake was curvilinear with a maximum at 3.5 percent of BW. Decreased estimated digestibility at lower intakes is counter to discounts in previous NRC versions, and the data set was limited in that range of intake, and many of the low DMI in the data set were from an experiment that fed low-quality (low digestibility) diets. Their data set also did not include any observations with DMI greater than about 5.5 percent of BW. Therefore, the committee decided to modify the equation to remove this depression at lower intakes while retaining the depression at higher intakes. This was done by calculating the marginal slope (first derivative of the DMI curve of de Souza et al., 2018) from a DMI of 3.5 percent of BW to the limit of the data (5.5 percent of BW) and averaging those values. The resulting average slope was 1.1, which was used as the linear discount factor for NDF as DMI (percentage of BW) increased.

On the basis of meta-data, White et al. (2017) derived an equation with a 7 percentage unit decrease in NDF digestibility per unit increase of DMI/BW. However, in their derivation, starch had no effect on NDF digestibility. In contrast, a meta-analysis of literature means by Ferraretto et al. (2013) estimated that increasing starch by 1 percentage unit linearly decreased NDF digestibility by 0.5 percentage units, but they reported DMI did not significantly affect NDF digestibility. In contrast with those meta-analyses, the negative effects of DMI and starch have been demonstrated in experiments specifically designed to test for those effects. Therefore, the committee adopted modified equations of de Souza et al. (2018), which include both a starch and DMI term (Equation 3-5a). Although this equation likely should include a factor to account for the fermentability of the starch, data were insufficient to do so.

$$\begin{aligned} \text{Digested proportion of NDF (dNDF_NDF)} \\ = \text{dNDF_NDF_base} - 0.0059 \times (\text{Starch_DM} - 26) \\ - 1.1 \times (\text{DMI_DM} - 0.035) \end{aligned}$$

(Equation 3-5a)

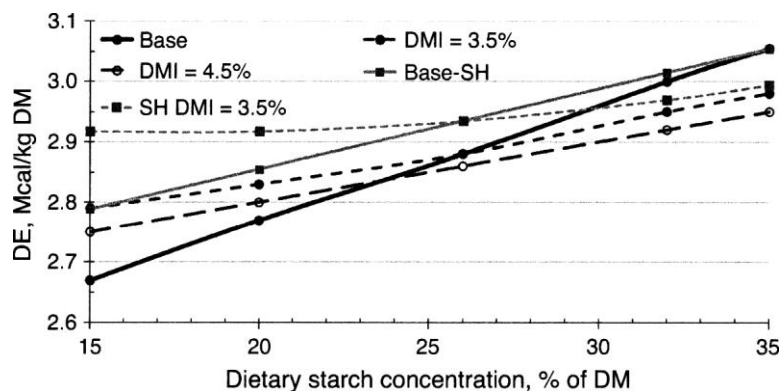


FIGURE 3-2 Effects of increasing dietary starch content on the DE value of example diets with and without adjusting NDF and starch digestibility for dietary starch concentration and DMI (Equations 3-5a and 3-5b) where starch replaces either forage NDF or soyhulls (SH). In all diets, the concentration of NDF + starch was 62 percent, and the diets contained 16 percent CP, 3 percent FAs, and 14 percent ROM with apparent digestibilities of 0.65, 0.73, and 0.70, respectively, and 5 percent ash. In one set of diets (black lines), base NDF digestibility and starch digestibility at 26 percent dietary starch and DMI of 3.5 percent of BW were 0.46 and 0.91 (typical values for a forage-based diet). The solid line represents the dietary DE content if no adjustments in NDF or starch digestibility were made for intake or dietary starch concentration. The line with short dashes shows DE content after adjusting for dietary starch concentration but keeping DMI at 3.5 percent of BW. The line with long dashes shows DE content after adjusting for dietary starch with a constant DMI of 4.5 percent of BW. In the second set of diets (gray lines), all concentrations and digestibilities were the same as above, except changes in dietary starch were achieved by exchanging starch with NDF of SH. The diet with 35 percent starch had no NDF from SH, but the diet with 15 percent starch had 20 percent NDF from SH. Assumed digestibility of NDF (without discount) from SH was 0.60 so that total diet NDF digestibility (without discount) ranged from 0.46 (35 percent starch and no SH) to 0.52 with 15 percent starch and 20 percent SH NDF. In both set of diets, note that the increased DE content expected from replacing NDF with starch is diminished by the drop in NDF digestibility caused by starch. Thus, if voluntary feed intake is not altered by diet, the value of added starch is less than expected. Also note that intake has less impact on DE content than does starch.

where starch is as a percentage of diet DM and DMI_BW = DMI/BW (kg/kg).

The digestibility of NDF in diets with substantial NDF from fibrous by-product feeds often decreases at a faster rate with increasing DMI than NDF in diets where most of the NDF is from long-forage particles (Potts et al., 2017; White et al., 2017); however, exceptions exist (Edionwe and Owen, 1989). The concentration of long-forage NDF likely affects the digestibility of NDF from shorter particles. With adequate long particles, small-particle NDF may be trapped in the rumen mat and be digested, whereas with inadequate long particles, the small particles flow from the rumen quicker without extensive digestion. Inadequate data are available to model these effects; therefore, particle size of the NDF source is not included in the model. The committee also recognizes that NDF digestibility could be depressed if diets contained inadequate RDP; however, data were deemed inadequate to derive an equation, and estimates are based on the assumption that RDP is not limiting, which is usually the case in practical situations.

Starch

In their 2018 article, de Souza et al. reported that starch digestibility decreases by 1.0 percentage unit for every 1-unit increase in DMI as a percentage of BW. Ferraretto et al. (2013) also found a negative relationship between DMI and starch digestibility, but the effect was 0.24 percentage units

per kilogram of DMI (approximately equal to a ~1.3 percent decrease per unit increase in DMI_BW). The data for both studies are biased heavily toward dry ground com grain, and it seems likely that digestibility of starch from more fermentable sources of starch (e.g., high-moisture com, barley, and wheat) would be affected less by intake; however, inadequate data were available to quantify a source of starch effect. In the model, when DMI_BW > 0.035 (i.e., 3.5 percent of BW), the proportion of starch that is digested decreases by 1.0 percent-age units per unit DMI_BW, and when DMI_BW < 0.035, the proportion of starch digested increases (Equation 3-5b). This adjustment may underestimate starch digestibility of highly fermentable starch sources when fed at high DMI.

$$\text{Digested proportion of Starch (dStarch_Starch)} \\ = \text{dStarch_Starch_base} - 1.0 \times (\text{DMI_BW} - 0.035)$$

(Equation 3-5b)

Examples of the effect of altering starch and NDF digestibility based on DMI and dietary starch concentrations on dietary DE are shown in Figure 3-2.

Other Considerations for Changes to the Base Digestibility

Almost no data exist to determine whether ROM digestibility is depressed with greater intake, but most components

of ROM are likely not affected by intake. As discussed above, digestibility of protein is not affected by DMI. The committee recognizes that the method of predicting effects of intake on digestibility in the current version was based on the inherent differences in ad libitum intake among cows or groups of cows sometimes fed diets of varying composition (e.g., Huhtanen et al., 2009; White et al., 2017; de Souza et al., 2018) rather than on designed experiments where intake was a treatment such as in studies by Tyrrell and Moe (1975). However, the first approach was considered more relevant for commercial applications where cows are generally fed ad libitum. Thus, these equations may not be accurate in situations where animals, especially heifers, are fed at restricted intake, and digestibility of NDF, starch, FA, and organic matter (OM) may be greater in restricted-fed animals than predicted by the current equations. However, de Souza et al. (2018) reported that their equations were reasonably accurate for predicting the digestion of NDF and starch in restricted-fed dairy heifers.

Estimating Endogenous Fecal Material and Apparent Digestibilities

The mass of fecal matter from endogenous sources cannot be measured in ruminants; it can only be estimated using statistical methods. This endogenous (or metabolic) fecal matter is mainly bacteria and bacterial residue comprising mostly CP and ROM. Endogenous CP and ROM are substantial and must be considered when converting truly digested nutrients into DE, which is calculated from apparent digestibility. In addition, these values are necessary for estimating the apparent digestibility values of CP and ROM for comparison to digestibility data collected in experiments. Apparent and true digestibilities are considered the same for NDF, starch, and FAs.

Endogenous fecal CP (to be consistent with terminology in Chapter 6, this will be referred to as metabolic fecal CP or MFCP), which represents sloughed endogenous cells and secretions, and undigested microbial CP are described in detail in Chapter 6.

$$\text{MFCP, g/kg DMI} = 11.62 + 0.134 \times \text{NDF_DMI} \quad (\text{Equation 3-6a})$$

where dietary NDF is as a percentage of DM.

$$\begin{aligned} \text{Fecal microbial CP (fMCP), g/kg DMI} \\ = (\text{Microbial CP (g/d)} \times 0.2) / \text{DMI} \end{aligned} \quad (\text{Equation 3-6b})$$

The endogenous masses were multiplied by the appropriate enthalpies (5.65 for CP and 4.0 Mcal/kg for ROM) to obtain endogenous fecal energy, which is subtracted from the sum of the DE from nutrients as discussed above. The DE values in the feed composition table (Chapter 19) were calculated with diet NDF set at 30 percent so that endogenous fecal CP = 15.6 g/kg DM or 0.088 Mcal/kg DMI. Undigested

bacterial CP was set as 16.5 g/kg DMI or 0.093 Mcal/kg (based on the average quantity of microbial protein synthesized and average DMI in the data set), and endogenous fecal ROM (efROM) was set at 34.3 g/kg DMI or 0.137 Mcal/kg DMI (Tebbe et al., 2017). Total endogenous fecal energy was $0.088 + 0.093 + 0.137 = 0.318$ Mcal/kg DMI.

To estimate apparent digestibilities of OM, CP, and ROM, which is useful in comparing model generated data to in vivo digestibility data, estimated endogenous output and estimated true digestibility of the fractions are used.

$$\begin{aligned} \text{Apparently Digested Proportion of ROM} \\ (\text{adROM_ROM}) = [(\text{ROM} \times 0.96) - 3.43] / \text{ROM} \end{aligned} \quad (\text{Equation 3-7a})$$

where ROM is a percentage of DM.

$$\begin{aligned} \text{Apparently Digested Proportion of CP (adCP_CP)} \\ = [(\text{RDP} + \text{dRUP}) - (\text{fMCP} + \text{MFCP})] / \text{CP} \end{aligned} \quad (\text{Equation 3-7b})$$

where all variables are kg/d.

$$\begin{aligned} \text{Apparently Digested Proportion of OM} \\ (\text{adOM_OM}) = (\text{NDF} \times \text{dNDF_NDF} + \text{Starch} \\ \times \text{dStarch_Starch} + \text{FA} \times \text{dFA_FA} + \text{RDP} + \text{dRUP} \\ + 0.96 \times \text{ROM} - \text{MFCP} - \text{fMCP} - \text{efROM}) / \text{OM} \end{aligned} \quad (\text{Equation 3-7c})$$

where NDF, Starch, FA, RDP, dRUP, ROM, OM, fMCP, MFCP, and efROM are in kg/d.

Estimating the Digestible Energy of Feeds and Diets

DE was calculated by multiplying the estimated truly digested nutrient concentrations for each feed by their respective heats of combustion and then subtracting the energy in the endogenous fecal excretions and undigested bacteria. To estimate DE from CP, the CP equivalent from supplemental NPN is subtracted from RDP. The NPN is assumed to have the energy value of urea, which is then added. As can be seen in Equation 3-1, ROM is corrected for supplemental NPN.

Digestible Energy

$$\begin{aligned} \text{Digestible Energy (DE_DM; Mcal/kg of DM)} \\ = 0.042 \times \text{NDF_DM} \times \text{dNDF_NDF} + 0.0423 \\ \times \text{Starch_DM} \times \text{dStarch_Starch} + 0.0940 \times \text{FA_DM} \\ \times \text{dFA_FA} + 0.0565 \times (\text{RDP_DM} \\ - \text{sNPNCP_DM} + \text{dRUP_DM}) + 0.0089 \\ \times \text{sNPNCP_DM} + 0.040 \times \text{ROM_DM} \times 0.96 - 0.00565 \\ \times \text{MFCP} - 0.00565 \times \text{fMCP} - 0.0040 \times \text{efROM_DM} \end{aligned} \quad (\text{Equation 3-8})$$

where feed fractions are a percentage of DM, endogenous fractions are g/kg, and digestibilities are expressed as proportions.

Estimating the Metabolizable Energy of Diets

Energy lost in urine and via methane is subtracted from DE to obtain ME. Gaseous energy (methane) is calculated as described in Chapter 14 as

$$\text{Gas Energy Loss (GasE_DM; Mcal/kg DMI)} = (0.294 \times \text{DMI} - 0.347 \times \text{FA_DM} + 0.0409 \times \text{dNDF_DM}) / \text{DMI}$$

(Equation 3-9)

where DMI is kg/d and FA_DM and dNDF_DM are a percentage of diet DM.

UE is calculated from estimated urinary N (N excretion):

$$\text{UN (g/d)} = \text{g Urinary N per day} = (\text{DMI} \times \text{CP_DM} \times \text{adCP_CP} - \text{Milk CP} - \text{Body gain CP}) \times 1,000 / 6.25$$

(Equation 3-10a)

where DMI, milk CP, and body gain CP are in kg/d, and CP_DM and adCP_CP are proportions.

If the animal is not lactating and within 60 days of parturition, the “Milk CP” term in Equation 3-10a is replaced with “0.00014 x Mature BW.” That term was derived by calculating the amount of protein retained (kg/d) in the gravid uterus at 250 days of gestation based on an average Holstein and Jersey calf birth weight (see gestation requirements in this chapter and in Chapter 6). For Equation 3-10a, body protein gain in lactating cows can be ignored because the effect is likely less than the imprecision associated with the equations. For example, a 100-g/d increase in body protein would change average estimated ME by <0.5 percent. For growing heifers, body protein gain is estimated as described in Chapter 11.

UE (Mcal/kg DMI) was estimated from urinary N excretion (g/d) as

$$\text{UE_DM (Mcal/kg DMI)} = (0.0146 \times \text{UN}) / \text{DMI}$$

(Equation 3-10b)

The coefficient (0.0146 Mcal/g of urinary N) was calculated from recent experiments that measured urinary energy and urinary N excretion (Morris et al., 2021).

Metabolizable energy was then calculated by subtracting gas and urinary energy from DE:

$$\text{ME_DM (Mcal/kg DMI)} = \text{DE_DM} - \text{GasE_DM} - \text{UE_DM}$$

(Equation 3-11)

In previous NRC editions, the conversion of DE to ME was considered on an individual feed basis. The current equation uses whole-diet estimates of urinary energy and gas energy, and thus it is only valid for the total diet. In NRC (2001), a correction was added to increase the efficiency of converting DE to ME for diets with higher fat so that the ef-

fective conversion for the fat was 100 percent. In the current equation, dietary FAs are used to estimate methane losses; therefore, the efficiency of converting DE to ME is greater for high-fat diets.

Estimating the Net Energy Lactation of Diets

The conversion of ME to NEL (Equation 3-12) is predicted on a whole-diet basis based on the average efficiency measured between 1974 and 1995 (see Table 3-2) in studies at the Beltsville Energy Metabolism Unit as reassessed by Moraes et al. (2015). The mean was 0.66 with a 95 percent confidence interval of 0.64 to 0.69. No correction for concentration of dietary fat is used with this equation:

$$\text{Net Energy of Lactation per kg DM (NEL_DM; Mcal/kg)} = 0.66 \times \text{ME_DM}$$

(Equation 3-12)

ENERGY REQUIREMENTS

Changes from the seventh revised edition include the following:

- The maintenance requirement is increased from 0.08 to 0.10 Mcal per kg of metabolic BW.
- The efficiency of using ME for lactation is increased from 0.64 to 0.66.
- Growth requirements have been simplified and are now explicitly related to the size of an animal relative to its mature BW (MatBW). The composition of gain does not change with diet and growth rate at a given BW, with the assumption that animals will be fed for rates of gain that maintain body condition.
- The composition of body condition score (BCS) change is not dependent on the starting BCS, so the energy required per kilogram of BW for body condition gain (or available from loss) is a constant.
- Requirements for physical activity have been updated.

The basic unit of dietary energy for dairy cattle is NEL, and all energy requirements are adjusted to be equivalent to this unit. Estimates for conversions of ME and changes in retained energy (RE) in the seventh edition (and several previous versions) were based on data from Moe et al. (1971) at the USDA Energy Metabolism Unit at Beltsville, Maryland. Recently, Moraes et al. (2015) reanalyzed the data from that laboratory, and the NEL requirement for maintenance was increased partly in response to this reanalysis and is biased toward the latest decade of work at Beltsville. Thus, the efficiency of using ME for replenishing body reserves and the efficiency of using mobilized body reserves for milk production also are now biased toward the later years of this reevaluation of the Beltsville data, as shown in Table 3-2.

TABLE 3-2 Energetic Parameters from Reanalysis of Data Over Several Decades from Beltsville Energy Metabolism Unit (Moraes et al., 2015) and Values for Seventh (NRC, 2001) and Current Editions^a

Parameter	1963-1995	1974-1995	Seventh Edition	Eighth Edition
ME for maintenance (Mcal/kg ^{0.75} BW/d)	0.14	0.16	0.13	0.15
NEL for maintenance (Mcal/kg ^{0.75} BW/d)	0.086	0.10	0.080	0.10
Conversion efficiencies				
ME to NEL	0.63	0.66	0.64	0.66
ME to RE during lactation	0.70	0.74	0.75	0.74
NEL to RE during lactation ^b	1.11	1.12	1.17	1.12
RE to NEL ^b	0.89	0.89	0.82	0.89
ME to RE when dry ^b			0.60	0.60
NEL to RE when dry ^b			0.94	0.91

^a All energy requirements must be converted to diet NEL equivalents for use in the model. NEL for maintenance is calculated as ME $\times$ conversion of ME to NEL. RE is retained energy, or the energy of tissue gain or loss.

^b The NEL required for RE during lactation or when dry is the conversion of ME to RE divided by the conversion of ME to NEL. Because ME is converted to tissue energy more efficiently than to milk energy, these values are greater than 1, so it takes less than 1 Meal of feed NEL to store 1 Meal of body tissue. The lower efficiency for dry cows likely is because dry cow diets are higher in fiber, resulting in greater heat of fermentation and diet-induced thermogenesis.

Maintenance Requirements

The NEL requirement for maintenance (NEL_{maint}) of adult dairy cattle is

$$\text{NEL}_{\text{maint}} (\text{Mcal/d}) = 0.10 \times \text{BW} \text{ kg}^{0.75} \quad (\text{Equation 3-13})$$

Based on Moraes et al. (2015), this value would have a 95 percent confidence interval of about ± 0.06 . This is a substantial increase from previous versions and adds about 2.5 Meal of NEL to the energy requirement of the average Holstein cow. Given the intensive selection for milk production in dairy cattle over the past 50 years with average milk production now three times that of the 1960s, it seems reasonable that modern dairy cows have metabolic rates for maintenance that are greater than they were 50 years ago. Cows of similar size and breed and in similar conditions may vary as much as 10 percent in their maintenance requirements (Van Es, 1961). This is consistent with more contemporary data from studies of residual feed intake showing that the intake for cows of similar BW and production varies by 7 percent after accounting for parity, location, diet, and other environmental effects (Tempelman et al., 2015); some of this variation could be caused by genetic variation in maintenance. Measured fasting heat production (Flatt et al., 1965) in dry nonpregnant dairy cows averaged 0.073 Mcal/unit metabolic BW (MBW), and estimated fasting heat production of dairy cows using regression analysis suggested an identical value (NRC, 2001). Because these measurements were made with cows housed in tie-stalls in metabolic chambers, past committees added a 10 percent activity allowance to account for normal voluntary activity of cows that would be housed in drylot or free-stall systems, such that the NEL_{maint} was set at 0.080 Mcal/kg MBW for mature dairy cows. This value has been used since NRC (1978). However, newer data and reevaluation of older

data all derived maintenance coefficients that were greater than 0.09, with some as high as 0.14. Moraes et al. (2015) reanalyzed the data from the Beltsville Energy Metabolism Unit and found that the apparent maintenance requirement for adult dairy cows increased with year of measurement. Maintenance requirements were 0.073, 0.087, and 0.122

Mcal/kg MBW for the years 1963 to 1973, 1974 to 1983, and 1984 to 1995, respectively, based on respective efficiencies of converting ME to NEL of 0.60, 0.62, and 0.69. Even with a lower efficiency of converting ME to NEL, the NEL_{maint} of cows from 1984 to 1995 would be greater than 0.10 $\times$ MBW. As with all requirements, maintenance requirements are not known with certainty. For example, assuming conversions of ME to NEL of 0.66, other studies have yielded NEL_{maint} (per kilogram of MBW) of 0.096 (Kirkland and Gordon, 1999), 0.09 (Birkelo et al., 2004), 0.11 (Xue et al., 2011), 0.11 (Dong et al., 2015), 0.14 (Foth et al., 2015), and 0.10 (Morris and Kononoff, 2021) for lactating cows; the NEM coefficient was 0.098 for fasted nonlactating cows (Birnie et al., 2000). The committee chose 0.10 $\times$ BW kg^{0.75} because it is simple and within the bounds determined by Moraes et al. (2015) for the last two decades of data in their study.

The most recent revision of the Nutrient Requirements for Beef Cattle (NASEM, 2016) also supports the higher value for dairy cattle. Converting their equation to BW (instead of shrunk BW) and adjusting for dairy breeds results in a NEL_{maint} of 0.095 $\times$ MBW. In addition, NASEM (2016) suggests that maintenance requirements per unit MBW do not decrease with age, they are 20 percent greater for lactating than nonlactating cows across beef breeds, and maintenance energy requirement is correlated positively with the genetic potential for milk production. Based on Table 19-1 of NASEM (2016), NEL_{maint} should be 0.095 $\times$ MBW for nonlactating and lactating dairy cows. The current committee recognizes that maintenance requirements could be considered greater for lactating than nonlactating cows because

(1) lactating cows generally have a greater mass of liver and other internal organs as a proportion of BW, (2) these organs produce more heat per unit mass than skeletal muscle, and (3) high-producing cows seem to require more ME for maintenance than low-producing cows (Moe et al., 1970; Baldwin et al., 1985; Ellis et al., 2006). However, the current committee considers that the increased mass and heat production of internal organs in lactating cows is a cost of milk production and should be assigned as part of the incremental heat loss in the conversion of ME to NEL for an animal that is digesting and metabolizing more feed nutrients.

The Committee on Nutrient Requirements of Beef Cattle (NASEM, 2016) applied a breed adjustment factor for maintenance of 1.2 for Holsteins and Jerseys (compared to British beef cattle breeds). Whether dairy cattle breed alters maintenance requirements or energy metabolism is not clear. Tyrrell et al. (1991) compared nonlactating and lactating Holstein and Jersey cows. Although actual milk yields (MYs) were greater for Holstein cows than for Jersey cows, energy output in milk as a function of MBW was similar, and there was no evidence to suggest that energy requirements for maintenance or production differed between breeds once adjusted for MBW. The committee considered setting maintenance requirement based on BW adjusted to a standardized BCS and to a nonpregnant status. Such an adjustment has been used for maintenance requirements in dogs and cats (Hand et al., 2000) and is consistent with the idea of setting maintenance energy requirements as a proportion of body protein mass (Agnew and Yan, 2000). Birnie et al. (2000) examined fasting heat production of 12 dry cows that were fed to be either thin or fat (mean BCS 1.3 versus 4.7 with BW of 467 versus 692 kg) and determined that daily NEL_{maint} per cow was essentially the same. The NEL_{maint} per unit MBW was also the same if BW was adjusted to a BCS of 3.0, assuming 1 BCS was 10 percent of BW. The committee recommends that future research examine the relationship between condition score and maintenance requirements but did not make any adjustment in the current requirement because most of the chamber data with dairy cows did not include information on body condition.

Lactation Requirements

The NEL concentration in milk is equivalent to the sum of the heats of combustion of individual milk components. No changes have been made to NEL requirements except for minor changes to equations for composition to account for true protein and NPN fractions. As with the seventh edition, the heats of combustion of milk fat, true protein, NPN CP equivalent, and lactose are 9.29, 5.71, 2.21, and 3.95 kcal/g, respectively. Milk CP, when estimated as $6.38 \times N$, contains 5 to 6 percent NPN (DePeters and Ferguson, 1992). Assuming milk CP is 6 percent NPN and 94 percent true protein, then the NEL value of milk CP is 5.5 kcal/g. If the CP content of milk is known and the true

protein content is not known, the NEL concentration of milk is calculated as

$$\text{NEL (Mcal/kg)} = 9.29 \times \text{kg Fat/kg Milk} + 5.5 \times \text{kg Crude Protein/kg Milk} + 3.95 \times \text{kg Lactose/kg Milk}$$

(Equation 3-14a)

If true protein is measured, the energy of true protein is adjusted up to account for the energy of NPN, which was assumed to equal 5.5 percent of milk CP and to have heat of combustion of urea (2.5 Mcal/kg) or $5.71 + 0.055 \times 2.5 = 5.85$. so the NEL concentration of milk is calculated as

$$\text{NEL (Mcal/kg)} = 9.29 \times \text{kg Fat/kg Milk} + 5.85 \times \text{kg True Protein/kg Milk} + 3.95 \times \text{kg Lactose/kg Milk}$$

(Equation 3-14b)

Milk lactose content is the least variable milk component and is generally about 4.85 percent of milk and varies only slightly with breed and milk protein concentration. If milk lactose is not measured, it should be set at 0.0485 kg/kg milk in the above equations.

When milk fat is the only milk constituent measured, NEL concentration can be calculated using the formula of Tyrrell and Reid (1965):

$$\text{NEL (Mcal/kg of milk)} = 0.360 + 0.0969 \times \text{Fat (\%)}$$

(Equation 3-14c)

The NEL system in this edition is based on yield of total energy in milk and does not account for many of the differences in metabolic transactions or the substrates required for synthesis of individual milk components. Attempts to assign differential efficiencies of converting feed ME to the NEL of individual milk components have been made (Baldwin, 1968; Dado et al., 1993); however, these calculations ignore energy losses in metabolic transactions outside of the mammary gland and thus are higher than those measured by calorimetry (Moraes et al., 2015). The measured calorimetric inefficiency of use of ME for milk includes losses associated with metabolic transactions for conversion of absorbed nutrients into milk components, the energy required for nutrient absorption, and increased rates of metabolism in visceral tissues required for support of increased milk production. Currently, data are lacking to confidently assign unique efficiencies of converting ME to the NEL of individual milk components.

Activity Requirements

The maintenance requirement is assumed to provide adequate energy for normal activity of cows in confinement. On many confinement farms, the distance between the housing area and milking center can be substantial, but this probably has little effect on overall energy expenditures

(see Chapter 13). Energy expended for walking over a level surface is approximately 0.35 kcal of NEL/kg of BW per kilometer walked (Brosh et al., 2006; Aharoni et al., 2009; Brosh et al., 2010). If the milking center was 200 m from the pen and a 650-kg cow was milked three times per day, the NEL expended for walking would be about 0.3 Mcal/d (about 2 percent of her maintenance requirement). Grazing cattle expend much more energy walking and gathering food; discussion and equations for grazing can be found in Chapter 13.

Environmental Effects

Equations for the energy requirements of thermal regulation have been developed and are used for beef cattle (NASEM, 2016). In general, lactating dairy cows typically operate at much higher metabolic activity level and produce more heat per day than do beef cattle. In addition, dairy cows are generally housed in environments that provide some shelter from cold conditions. Thus, cold stress is not as important for dairy cows as beef cattle. For lactating cows in cold environments, the change in energy requirement is probably minimal because of the normally high heat production of cows consuming large amounts of feed, and they likely require very little extra dietary energy to counteract cold environments if they are kept dry and are not exposed directly to wind. Young (1976) summarized experiments with ruminants in which an average reduction in DM digestibility of 1.8 percentage units was observed for each 10°C reduction in ambient temperature below 20°C. Much of this lowered digestibility under cold stress was related to an increased rate of passage through the digestive tract (Kennedy et al., 1976). Because of the effects of low temperature on digestibility, under extremely cold weather conditions, feed energy values could possibly be lower than expected.

Dairy cows are often heat-stressed, and lactating cows producing the most milk are the most likely to be heat-stressed. However, the committee determined that insufficient data were available to quantify these effects accurately in dairy cows and to account for all important factors such as ambient temperature, relative humidity, radiant energy exposure, night-cooling, air speed, level of production, and heat abatement programs. Heat stress may increase the maintenance requirement of dairy cattle by 7 to 25 percent (NRC, 1981), but these values are based on very little direct data. Measured by indirect calorimetry, fasting heat production increased about 5 percent and ME requirement for maintenance increased about 10 percent when dry cows were housed at 36°C compared with 18°C (Kurihara, 1996). Heat stress induces behavioral and metabolic changes in cattle (West, 1994; Wheelock et al., 2010). Some changes, such as increased respiration rate, panting, and immune activation, likely increase energy expenditures, but the most important responses to heat stress are the physiological responses that result in decreased milk production and feed consumption. When cows

eating ad libitum in thermoneutrality (temperature humidity index of 75) were either heat-stressed (temperature humidity index of 65) or pair-fed at thermoneutrality, heat production per unit of MBW was decreased similarly, likely due to decreased feed intake, and no increase in the maintenance requirement was detected (Lamp et al., 2015). Heat-stressed animals employ novel homeorhetic strategies that decrease milk production and decrease feed intake without a change in lipid mobilization (Baumgard and Rhoads, 2013). The decrease in DMI induced by heat stress is greater than what would be expected based on the decrease in MY therefore, equations developed to estimate DMI under thermoneutral conditions will likely overestimate DMI under heat stress, and users may need to modify the DMI estimates. Ultimately, these changes decrease the need for heat dissipation and are important for survival. However, they are difficult to model. Because of limited data, adjustments for heat stress have not been included in the calculation of maintenance requirements; further research is needed.

Pregnancy Requirements

Energy requirements for gestation in NRC (2001) were calculated from a linear function of day of gestation starting at day 190 and scaled to calf birth weight based on serial slaughter data (Bell, 1995; Bell et al., 1995). However, over a longer gestation period, the gravid uterine growth is better described by a logistic or decaying exponential growth function (Koong et al., 1975; Ferrell, 1991). The function was rearranged so that birth weight of the calf was an input rather than an output. Gravid uterine weight at parturition ($GrUter_Wt_{(t = \text{parturition})}$) and uterine weight immediately after calving ($Uter_Wt_{(t = \text{Parturition})}$) were estimated from calf birth weight using data from Bell et al. (1995) and House and Bell (1993) data:

$$GrUter_Wt_{(t = \text{parturition})} = \text{Calf birth weight} \times 1.825 \quad (\text{Equation 3-15a})$$

$$Uter_Wt_{(t = \text{Parturition})} = \text{Calf birth weight} \times 0.2288 \quad (\text{Equation 3-15b})$$

Average birth weight (kg) of calves born from multiparous cows are 44 (Holstein), 26 (Jersey), 38 (Ayrshire), 48 (Brown Swiss), 36 (Guernsey), and 36 (milking shorthorns), and birth weight of calves born from heifers averages 91 percent of those weights (Legault and Touchberry, 1962; Olson et al., 2009; Dhakal et al., 2013; Kamal et al., 2014). Calf birth weight also can be estimated from MatBW: 0.063 times MatBW for a cow and 0.058 times MatBW for a heifer. For these calculations, full term is assumed to be 280 days of gestation, which is also used in the software.

Nonlinear regression (i.e., a logistic function) of the data from Bell et al. (1995) and House and Bell (1993) was used to derive Equation 3-16a. The model also predicts uterine

involution postpartum (Equation 3-16b) to maintain mass balance and predict release of tissue energy and amino acids for productive use in early lactation as described by Hanigan et al. (2009):

$$\text{GrUter_wt} = (\text{GrUter_Wt}_{(t = \text{parturition})} \times e^{-(0.0243 - (0.0000245 \times \text{DayGest})) \times (280 - \text{DayGest})})$$

(Equation 3-16a)

$$\text{UterWt} = ((\text{Uter_Wt}_{(t = \text{Parturition})} - 0.204) \times e^{-0.2 \times \text{DayLact}}) + 0.204 \quad (\text{Equation 3-16b})$$

where DayGest = day of gestation (day of gestation must be

between 12 and 280), and $\text{Uter_Wt}_{(t = \text{parturition})}$ = estimated

weight (kg) of uterus immediately postcalving (Calf birth-weight $\times 0.2288$). The involution rate is not known with certainty, but the value of 0.2/d will result in essentially complete involution by day 21 of lactation.

Daily rates of wet tissue deposition (kg/d) are derived from Equations 3-16a and 3-16b as (variables defined above):

$$\text{During gestation: GrUter_Wt}_{\text{Gain}} = (0.0243 - (0.0000245 \times \text{DayGest})) \times \text{GrUter_Wt}$$

(Equation 3-17a)

$$\text{During involution: GrUter_Wt}_{\text{Gain}} = -0.2 \times \text{DayLact} \times (\text{Uter_Wt} - 0.204) \quad (\text{Equation 3-17b})$$

The NEL gestational (Gest) requirements were calculated from the rate of change in gravid uterine tissue mass and by assuming tissue contained 0.882 Mcal of energy/kg (House and Bell, 1993; Bell et al., 1995), an ME to gestation energy efficiency of 0.14 (Ferrell et al., 1976; NRC, 2001), and an ME to NEL efficiency of 0.66.

$$\text{Gest_NEL (Mcal/d)} = \text{GrUter_Wt}_{\text{gain}} \times (0.882/0.14) \times 0.66 = \text{GrUter_Wt}_{\text{gain}} \times 4.16$$

(Equation 3-18)

Over a 60-day dry period, NEL requirements are essentially the same whether calculated using NRC (2001) or the new

model; however, calculated NEL requirements will be lower for far-off dry cows and greater in prefresh cows using the new model as compared to the previous model (see Table 3-3). Protein is discussed in Chapter 6.

Changes in Body Weight and Composition During Growth and Lactation

In the seventh edition, body composition equations were based largely on data from beef cattle with the standard reference animal having a MatBW of 500 kg. Modern Holsteins have a MatBW of ~700 kg (Tempelman et al., 2015), and dairy breeds are generally less muscular than beef breeds.

Several publications have reported the composition of growing and mature Holsteins in the past 20 years, and the com-

mittee deemed that sufficient data were available to develop equations for Holsteins. Details on the data set and models can be found in de Souza et al. (2018).

In this edition, body energy change is partitioned as (1) body frame gain (i.e., true growth), (2) body reserves or condition gain (or loss), and (3) pregnancy-associated gain (considered a pregnancy requirement). Frame gain is normal skeletal growth, the normal body gain that occurs as animals mature from birth to adult, and includes the normal gains in skeletal muscle, adipose tissue, bone, organs, intestinal tract, and gut contents. Frame gain is the gain in BW without overnight fasting, assuming an animal maintains a constant BCS and is not pregnant. All requirements in the current model related to growth assume BW gain is frame gain. The tissue that is lost and gained during times of nutrient excess or deficiency in the life of an animal is body reserves. Changes in body reserves are generally, but not always, observed as changes in BCS. Condition score changes are expected during a normal lactation cycle but can also occur in growing heifers if fed more or less than needed for normal growth. Pregnancy-associated gain includes the growing fetus and associated tissues, including placenta and mammary gland that increase as gestation progresses and are considered a pregnancy requirement.

BW can be divided into empty BW (EBW, the actual tissues of the animal) and gut fill. Based on data of lactating dairy cows, gut fill is about 5.2 times DMI (Gibb et al., 1992;

TABLE 3-3 Comparison of Gestation Energy and Protein Requirements^a Calculated Using NRC (2001) and Current Model (Assumed Birth Weight of Calf=44 kg)

Day of Gestation	Gestation NEL, Mcal/d		Gestation MP, g/d	
	NRC (2001)	Current	NRC (2001)	Current
50	0	0.04	0	3
100	0	0.1	0	13
150	0	0.5	0	43
200	2.7	1.4	199	125
220	3.0	2.0	245	185
250	3.4	3.5	306	320
275	3.8	5.4	357	489

^a See Chapter 6 for the details on protein.

Andrew et al., 1994). For the typical cow, eating at 3.5 percent of BW, gut fill would be 18 percent (5.2 x 3.5 percent) of BW similar to NRC (2001) and NASEM (2016). Thus, 1 kg of frame gain for a cow includes 0.82 kg of EBW and 0.18 kg of gut fill. The committee recognizes that gut fill may not be 18 percent of BW gain in all cases, especially if animals are fed at restricted intake or fed diets of mostly poor-quality forage.

Energy of Tissue Mobilization and Repletion

The tissue that is lost and gained during a lactation cycle or during other times of nutrient deficiency or excess in the life of a cow is mostly lipid and considered body energy reserves. Like most mammals, a dairy cow typically mobilizes body reserves during early lactation and repletes them during later lactation and the dry period. Optimum management of body reserves improves the health and profitability of dairy cows. Overly fat cows, especially those around the time of calving, have lower feed intake and increased risk for dystocia and health problems. Conversely, overly thin cows have insufficient reserves for maximum milk production and often do not conceive in a timely manner.

Changes in body energy reserves are usually observed as changes in BCS. Although evaluation of BCS is subjective in nature, it is the only practical method to evaluate body energy stores of dairy cows on most farms. In the United States, the most common systems of BCS use a 5-point scale originally proposed by Wildman et al. (1982) with a BCS of 1 being extremely thin and a score of 5 being extremely fat. Edmonson et al. (1989) developed a BCS system using a

5-point scale based on visual appraisal of eight separate body locations (see Figure 3-3). Analysis of variation due to cows and to individuals assessing BCS suggested that visual appraisal of just two locations (between the hooks and between the hooks and pins) had the smallest error due to assessor and accounted for the greatest proportion of variation due to individual cows.

Despite the emphasis on measuring BCS over that past 30 years, data are surprisingly lacking on the mathematical relationships between BCS, BW change, gut fill, and body composition changes of dairy cows. Much of the available data are from transition cows during which time BCS and feed intake are changing in opposite directions so that actual BW loss is masked by increases in gut fill as feed intake increases during early lactation. Studies are needed in this area.

Body Weight Change per Body Condition Score

In NRC (2001), each BCS unit was associated with a change in BW of ~14 percent, or about 80 kg for a typical Holstein cow, and the weight gain or loss associated with changes in BCS was considered to be 18 percent gut fill.

Using deuterium oxide dilution, Komaragiri and Erdman (1997) observed a change of 63 kg per unit BCS in cows with an average BW of 667 kg, and Komaragiri and Erdman (1998) observed a change of 59 kg per unit BCS in cows with an average BW of 634 kg. Other studies found BW per BCS values of 56 kg for 640-kg cows (Chillard et al., 1991), 56 kg for 558-kg cows (Otto et al., 1991, which was incorrectly interpreted in the seventh edition), and 56 kg for 597-kg cows

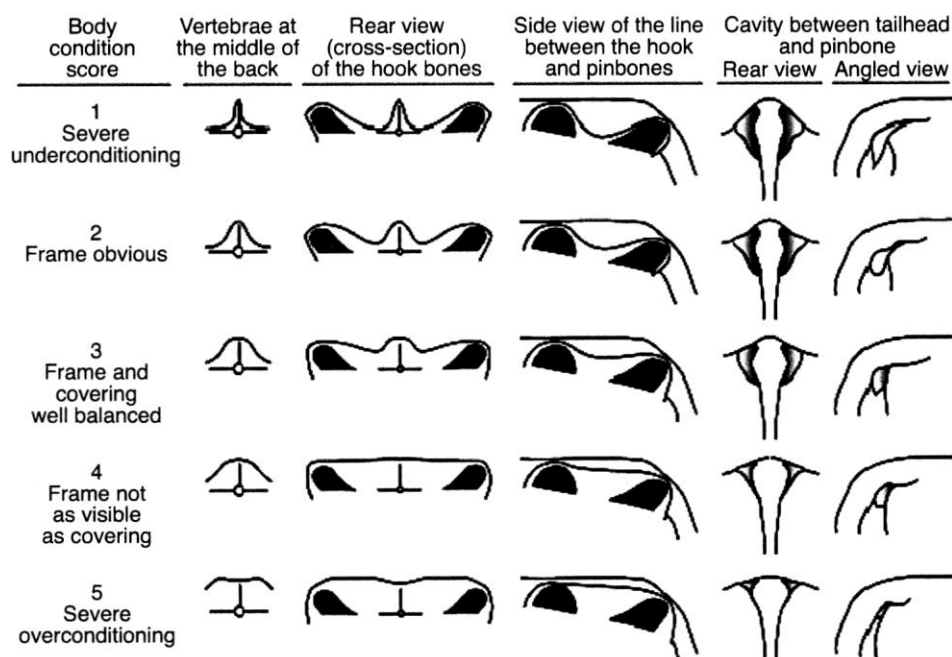


FIGURE 3-3 Body condition scoring chart.

SOURCES: M'hamdi et al. (2012); adapted from Edmonson et al. (1989).

(Waltner et al., 1994). A summation of the data across these studies suggests a mean BW change per unit of BCS of 9.4 percent of BW. This BW change is assumed to be entirely EBW change; in other words, changes in body mass that are due to gains or losses in BCS are not associated with changes in gut fill. In the seventh edition, gut fill was set at 18 percent of BW, and therefore, the mass of gut fill changed with BW as BCS was lost or gained in lactating cows. In growing animals, gut fill increases proportionally as an animal matures, but this is not true for cows, in which gut fill varies with the changes in DMI (Andrew et al., 1994) during the lactation cycle. Therefore, changes in BW associated with changes in BCS are assumed to be all body tissue (EBW) with no change in gut fill per unit BCS; hence, 1 kg of live body gain is 1 kg of empty body gain for changes in BCS. Cows do not eat more as they gain BCS; in fact, BCS is inversely associated with DMI (Garnsworthy, 2006; de Souza et al., 2019). Assuming gut fill is 18 percent of BW, a change in 1 BCS unit would be equal to 11.5 percent of BW for a cow at a BCS of 3. This value should be slightly higher for a thin cow and slightly lower for a fat cow.

Composition and Energy Content of Changes in Body Reserves

In the seventh edition, the composition of changes in body reserves was dependent on the starting and ending BCSs, with a greater proportion of fat in the change as average BCS increased. With that system, the RE of empty body changes associated with reserves varied from 5.1 Mcal/kg in very thin cows to 9.6 Mcal/kg in very fat cows. However, based on the constant fat content per unit BW change cited earlier, this is not supported by evidence. The current committee deemed the difference too small to warrant the increased complexity of differential energy values for BCS changes for cows with BCS ranging from 2 to 4. Most cows on farms are within these bounds; hence, the composition of BW change for BCS changes is considered a constant.

The energy value of a kilogram of true body tissue that is lost or gained is dependent on the relative proportions of fat and protein in the tissue and their respective heat of combustion. As in the seventh edition, the committee chose 9.4 and 5.55 Mcal/kg for retained body fat and protein. The current committee estimates that gain or loss of empty body in lactating cows between BCS of 2 and 4 contains 62.2 percent fat, 27.6 percent water, 8.1 percent protein, and 2.1 percent ash and has an energy value of 6.3 Mcal/kg. These values are based on the fat content of EBW from Chillard et al. (1991), Komaragiri and Erdman (1997, 1998), Otto et al. (1991), and Waltner et al. (1994), as well as the protein and ash content of fat-free mass of Waldo et al. (1997). Because gut fill does not change with BCS, the composition and energy value of BCS gain or loss is the same on a BW as an EBW basis. Assuming that 1 BCS unit equals 9.4 percent of BW, a 1-unit change in BCS for a 650-kg cow equals 61 kg of body mass containing 385 Meal of energy and 5.0 kg of

protein. Because ME is used more efficiently for gain of reserves than for production of milk during lactation (0.74 versus 0.66), a 1-unit gain in BCS (385 Meal of reserve RE) requires 520 Meal of ME, and this equates to only 343 Meal of feed NEL. Conversely, a loss of 1 BCS unit for a 650-kg cow would equal a loss of 385 Meal of RE and provide 343 Meal of NEL (equivalent to the energy in 490 kg of milk with 3.5 percent fat).

Based on the values from Table 3-2, the energy requirement in NEL units for body reserves gain is as follows:

If lactating:

$$\begin{aligned}\text{NEL (Mcal/kg gain)} &= 6.3 \text{ Meal RE/kg} \times 0.89 \\ &= 5.6 \text{ Meal NEL/kg BW gain}\end{aligned}\quad (\text{Equation 3-19a})$$

If not lactating:

$$\begin{aligned}\text{NEL (Mcal/kg gain)} &= 6.3 \text{ Meal RE/kg} \times 1.10 \\ &= 6.9 \text{ Meal NEL/kg BW gain}\end{aligned}\quad (\text{Equation 3-19b})$$

The NEL available from mobilization of body tissue and thus not needed in the diet is

$$\begin{aligned}\text{NEL available (Mcal/kg loss)} &= 6.3 \text{ Meal RE/kg} \\ &\times 0.89 = 5.6 \text{ Meal NEL/kg BW loss}\end{aligned}\quad (\text{Equation 3-19c})$$

Mobilization of body tissue is normal during early lactation to support the energy needs for lactation, as it is in many mammalian species. A loss of 0.5 BCS units typically occurs during the first 60 days postpartum in dairy cows.

Energy Requirements for Frame Growth

In NRC (2001), energy requirements for growth were developed for heifers using the beef NRC (1996) system. The RE associated with gain was dependent on where the animal was in its growth curve relative to a standard reference animal with 498 kg MatBW and on the animal's average daily gain (ADG). The effect of ADG on the composition of gain was very small, with ADG taken to a power of 1.097. Published reports in the past 20 years with widely divergent nutritionally induced changes in rate of gain of Holstein heifers show that the composition of gain can change much more than that previous equation estimated (Radcliff et al., 1997; Brown et al., 2005; Meyer, 2005; Davis Rincker et al., 2008). In studies where BCS was measured, diets that cause different rates of gain can cause large differences in BCS (Radcliff et al., 1997); however, in studies with younger heifers, diets that cause divergent rates of gain resulted in very little change in the composition of gain (Meyer, 2005). Ideally, the changes in body composition due to fast or slow daily gain from dietary manipulation of heifers should be assigned to changes in body

reserves and not to frame gain, but data on the effects of diet on gain and BCS are lacking. Thus, due to insufficient data, no allowance was made for rate of gain to alter the fat content of growing heifers in the current model. For growing heifers, BCS is not used, but the assumption is that the heifers will be fed to maintain moderate body condition. The committee recommends that further studies be conducted so that body gain of heifers can eventually be partitioned into frame gain and condition change, using a system to assess change in body fatness such as body condition scoring or ultrasonic fat depth.

The committee reemphasizes that frame gain assumes appropriate gains of lean and fat tissues for an animal maintaining a BCS of 3. An animal can gain frame mass while losing body condition. The seventh edition allowed for cows in their first and second lactations to gain frame mass and change condition simultaneously, but the growth equations for cows were not included in the computer model. The current version supports both frame gain and body condition changes for cows but includes only frame growth for heifers.

In the current version, both BW and the gain in BW for frame growth in heifers are 85 percent tissue and 15 percent gut fill. For immature cows, gut fill is calculated as 18 percent of BW or BW gain. Requirements for frame growth are described and justified in Chapter 11. The equations are as follows:

$$\text{Fat in Frame ADG (Fat_ADG), g/g} = (0.067 + 0.375 \times (\text{BW/MatBW})) \times \text{EBG/ADG} \quad (\text{Equation 3-20a})$$

$$\text{Protein in Frame ADG (Protein_ADG), g/g} = (0.201 - 0.081 \times (\text{BW/MatBW})) \times \text{EBG/ADG} \quad (\text{Equation 3-20b})$$

$$\text{RE of Frame ADG (RE_FADG), Mcal/kg} = 9.4 \times \text{Fat_ADG} + 5.55 \times \text{Protein_ADG} \quad (\text{Equation 3-20c})$$

The efficiency of converting feed ME to net energy for gain (NEg) using NRC (2001) equations averaged about 0.40. The efficiency of converting NEL to NEg is based on conversions of 0.40 for ME to NEg and 0.66 for ME to NEL and is thus $0.40/0.66 = 0.61$; therefore,

$$\text{ME for Frame ADG (ME_FADG, Mcal/kg)} = \text{RE_FADG}/0.4 \quad (\text{Equation 3-20d})$$

$$\text{NEL for Frame ADG (NEL_FADG, Mcal/kg)} = \text{RE_FADG}/0.61 \quad (\text{Equation 3-20e})$$

Comparison of New Energy System to the 2001 System

A variety of diets that differed in forage quality and concentrations of starch, forage NDF, total NDF, fat, and CP fed at DMIs of 3.5 and 4.8 percent of BW were evaluated using

the NRC (2001) equations and equations in this version. Two dry cow diets were also evaluated. The concentration of NEL (Mcal/kg) in lactating cow diets averaged about 8 percent higher (range, 6 to 12 percent) using the new model, and dry cow diets were about 10 percent higher. Using the 2001 model, energy concentrations in the lactating cow diets ranged from 1.53 to 1.63 Mcal/kg and from 1.65 to 1.77 Mcal/kg for the new model. Concentrations of NEL in dry cow diets increased from about 1.4 to 1.55 Mcal/kg. The greatest difference for lactating cow diets was observed at the high intake (about 10 percent higher). Generally, the NEL concentration of high-starch diets increased more than low-starch diets when comparing the new model to the old model. Energy requirements on average increased about 8 percent with the greatest relative increase for high-BW, low-producing cows. Energy requirements using the new system increased about 6 percent for a 650-kg cow producing 55 kg of milk but by about 11 percent for the same cow producing 30 kg of milk compared to NRC (2001). The greatest relative effect was on dry cows. The energy requirement of a 700-kg dry cow 20 days before calving increased by about 30 percent. Because this is an energy system, the comparison that is most important is energy balance ($\text{NEL intake} - [\text{NEL for maintenance} + \text{lactation} + \text{gestation} + \text{growth}]$). For lactating cows, NEL balance averaged about 0.6 Mcal/d (about 1.5 percent of NEL requirement) more with the new system compared to NRC (2001). For dry cows, NEL balance was about 2.8 Mcal less with the new system compared to the old one. For lactating cows, the difference between the new system and the old system was related to milk production. Net energy balance was less with the new system for lower-producing cows than the old system. Conversely, NEL balance was greater with the new system than the old system for higher-producing cows. This means that a higher-energy diet fed to a high-producing cow (also high DMI) will support greater milk production using the current system than the same diet would using the NRC (2001) system. Conversely, using the current system a cow would need to consume a slightly higher-energy diet to obtain the same production and body condition than would the NRC (2001) system. For dry cows, a diet formulated to exactly meet requirements using NRC (2001) would not meet the energy requirements using the current system even though energy density of the diet would be greater with the current system than with NRC (2001).

Energy Partitioning

Production response to increased energy intake is dependent on how energy is partitioned between MY and body energy reserves. Energy partitioning is mostly affected by stage of lactation but also by the interaction between diet and the physiological state of cows as they progress through lactation. Cows that produce more milk need more glucogenic fuels, so increasing the starch content of rations results in a more positive milk response for cows that produce more milk.

(Voelker et al., 2002). MY response to dry ground corn substituted for soyhulls at 30 percent of the diet DM increased linearly with MY as it increased from 28 to 62 kg/d, with no response for cows at the lower end of the range in MY (Boerman et al., 2015). As lactation proceeds, insulin concentration and sensitivity of tissues increase. Increasing glucose supply beyond that required for milk production increases plasma concentrations of glucose and insulin and partitioning of energy to body reserves. Intravenous glucose infusion of up to 30 percent of NEL requirement in mid-lactation cows linearly increased plasma insulin concentration, energy balance, BW, and back fat thickness, without affecting DMI or MY (Al-Trad et al., 2009).

Decreasing diet starch content by substitution of high-fiber by-products, or even fat, for cereal grains increases energy partitioning to milk (Boerman et al., 2015; Potts et al., 2017). For instance, substitution of soyhulls for dry ground corn in diets of mid-lactation cows increased yield of milk fat linearly with a subsequent linear decrease in BW with no effect on MY (Ipharraguerre et al., 2002). In addition, substitution of beet pulp for barley grain in rations fed to cows in late lactation linearly decreased plasma concentrations of glucose and insulin, BCS, and back fat thickness; increased ruminal pH linearly; and tended to linearly increase milk fat yield and milk energy output (Mahjoubi et al., 2009).

Increasing dietary starch can increase the risk of milk fat depression by altering ruminal biohydrogenation of long-chain unsaturated FAs (Bauman et al., 2011), as discussed in Chapter 4. Certain conjugated linoleic acid (CLA) isomers, including trans-10, cis-12 C18:2, are produced in the rumen when biohydrogenation is altered by highly fermentable diets. This CLA isomer downregulates several genes involved in lipogenesis, decreasing de novo FA synthesis in the mammary gland (Baumgard et al., 2002) while having opposite effects on expression of genes involved in lipogenesis in adipose tissue (Harvatine et al., 2009; Jenkins and Harvatine, 2014). Thus, this CLA isomer has a role in energy partitioning by reducing milk energy output sparing energy for lipid synthesis in adipose tissue.

Feeding high-starch diets to high-producing cows in early lactation may support maximal production of milk with minimal loss of body reserves; however, in later lactation, once cows have adequate body reserves, replacing starch with other energy sources such as digestible fiber can help prevent overfattening while still maintaining high milk production. Although a quantitative prediction of effects of diet on energy partitioning is not currently feasible, the effects of diet on partitioning should be considered when formulating diets and are useful when combined with observation of cow responses to diets on farms.

Feed Efficiency

Feed efficiency is a complex trait for which no single definition is adequate. For simplicity, dairy feed efficiency

is usually defined as milk output per unit of feed input, with the units generally being mass, energy, protein, or economic value. Although the major product for a dairy cow is milk, changes in body tissue can result in misleading values for feed efficiency and should not be ignored. When evaluating feed efficiency over an animal's lifetime, all feed used as a calf, heifer, and cow and all products produced, including milk, meat, and newborn calves, should be considered. When evaluating feed efficiency of lactating cows for portions of a lactation, corrections should be made for changes in body tissue as

$$\text{Feed efficiency} = \frac{\text{Milk energy} + \text{Change in body energy}}{\text{Feed energy input}} \text{ (Equation 3-21)}$$

Feed efficiency could also account for feed that is wasted by the cow and losses that occur during harvesting, storing, mixing, or feeding. To define efficiency on a global scale, consideration should be given to human-consumable inputs versus other foods, fossil fuels, water, and land, as well as outputs of greenhouse gases, pollutants, fertilizers, and other products not used for human consumption. How dairy cattle are fed also impacts the broader ecosystem rural sociology, food quality, animal well-being, the need for oil, and the beef industry (fewer dairy cows will increase the need for beef cows). These considerations have been discussed (Oltjen and Beckett, 1996; Arriaga et al., 2009; Capper and Bauman, 2013; Connor, 2015; VandeHaar et al., 2016). Improvements in feed efficiency generally translate into improvements in environmental sustainability, as illustrated by Capper et al. (2009).

Feed efficiency, no matter which metric is used, is generally greater with greater milk production per cow (VandeHaar et al., 2016). The first portion of feed eaten by a cow is used for maintenance; feed consumed above maintenance requirement is captured in milk or tissue. If milk energy output is considered in units needed for maintenance, then a cow producing milk at 3x her maintenance requirement uses only 25 percent of her NEL intake for maintenance and can use 75 percent for milk, assuming no change of body tissue. At 4x maintenance, she uses 80 percent of her NEL intake for milk. The dairy industry in North America has increased feed efficiency considerably over the past 100 years as milk production has increased. Currently, the average cow operates at ~3x maintenance intake, so there is still room for improvement. VandeHaar (1998) estimated that Holsteins with a MatBW of 625 kg would attain nearly maximal lifetime efficiency at 21,000 to 24,000 kg of milk per year. At one time, this seemed an unlikely level of productivity for the average farm, but with current technologies, it now seems possible. However, because maintenance requirements per unit of MBW have increased, higher levels of production will be needed to achieve maximal efficiency. If maintenance is 25 percent greater (0.10 versus 0.08), then the milk production to achieve maximal efficiency also will be 25 percent

greater, assuming no change in MatBW. To continue to improve efficiency in dairy cattle, the industry may need to focus more on efficiency as a goal than as the by-product of focusing on productivity. Breeding programs have started to focus on efficiency by selecting against larger cows and by selecting for a more negative residual feed intake, which is a measure of actual versus predicted intake for an individual cow. Residual feed intake is not very useful in making nutrition and management decisions on farms, but it shows promise as a tool for genetic selection (Veerkamp et al., 1995; Connor, 2015; Pryce et al., 2015; Tempelman et al., 2015).

When feeding and managing cows, maximizing feed efficiency, as defined by milk output per unit feed input, is seldom a worthy goal. Diets high in fat, starch, and protein and low in fiber will almost always increase milk to feed ratio, but these types of diets are not always conducive to optimal profit, health, and sustainability. As described earlier in this chapter and elsewhere, high-grain (starch) diets are more digestible and can increase feed intake and milk solids output during peak lactation. However, high starch decreases digestibility of fiber, and high starch and fat can decrease feed intake in some cases. Monitoring responses to diets is a key part of managing for efficient milk production. Moreover, one of the important contributions of ruminants is their ability to digest foods that humans cannot effectively use or will not consume. Cattle can make use of fiber and thus enable humans to indirectly derive nutrients from fiber. Ruminants can convert the myriad of high-fiber by-product feeds that are available across most of the world into human food.

REFERENCES

- Agnew, R. E., and T. Yan. 2000. Impact of recent research on energy feeding systems for dairy cattle. *Livestock Prod. Sci.* 66:197-215.
- Aharoni, Y., Z. Henkin, A. Ezra, A. Dolev, A. Shabtay, A. Orlov, Y. Yehuda, and A. Brosh. 2009. Grazing behavior and energy costs of activity: A comparison between two types of cattle. *J. Anim. Sci.* 87(8):2719-2731.
- Aikman, P. C., C. K. Reynolds, and D. E. Beever. 2008. Diet digestibility, rate of passage, and eating and rumination behavior of Jersey and Holstein cows. *J. Dairy Sci.* 91:1103-1114.
- Al-Trad, B., K. Reisberg, T. Wittek, G. B. Penner, A. Alkaassem, G. Gabel, M. Furl, and J. R. Aschenbach. 2009. Increasing intravenous infusions of glucose improve body condition but not lactation performance in midlactation dairy cows. *J. Dairy Sci.* 92:5645-5658.
- Andrew, S. M., D. R. Waldo, and R. A. Erdman. 1994. Direct analysis of body composition of dairy cows at three physiological stages. *J. Dairy Sci.* 77:3022-3033.
- Arriaga, H., M. Pinto, S. Calsamiglia, and P. Merino. 2009. Nutritional and management strategies on nitrogen and phosphorus use efficiency of lactating dairy cattle on commercial farms: An environmental perspective. *J. Dairy Sci.* 92:204-215.
- Bal, M. A., J. G. Coors, and R. D. Shaver. 1997. Impact of the maturity of corn for use as silage in the diets of dairy cows on intake, digestion, and milk production. *J. Dairy Sci.* 80:2497-2503.
- Baldwin, B. R., N. E. Forsberg, and C. Y. Hu. 1985. Potential for altering partition in the lactating cow. *J. Dairy Sci.* 68:3394-3402.
- Baldwin, R. L. 1968. Estimation of theoretical calorific relationships as a teaching technique. *J. Dairy Sci.* 51:104.
- Bauman, D. E., K. J. Harvatine, and A. L. Lock. 2011. Nutrigenomics, rumen-derived bioactive fatty acids, and the regulation of milk fat synthesis. *Annu. Rev. Nutr.* 31:299-319.
- Baumgard, L. H., and R. P. Rhoads. 2013. Effects of heat stress on postabsorptive metabolism and energetics. *Annu. Rev. Anim. Biosci.* 1:311-337.
- Baumgard, L. H., B. A. Corl, D. A. Dwyer, and D. E. Bauman. 2002. Effects of conjugated linoleic acids (CLA) on tissue response to homeostatic signals and plasma variables associated with lipid metabolism in lactating dairy cows. *J. Anim. Sci.* 80:1285-1293.
- Bell, A. W. 1995. Regulation of organic nutrient metabolism during transition from late pregnancy to early lactation. *J. Anim. Sci.* 73:2804-2819.
- Bell, A. W., R. Slepatis, and R. A. Ehrhardt. 1995. Growth and accretion of energy and protein in the gravid uterus during late pregnancy in Holstein cows. *J. Dairy Sci.* 78:1954-1961.
- Birkelo, C. P., M. J. Brouk, and D. J. Schingoethe. 2004. The energy content of wet corn distillers grains for lactating dairy cows. *J. Dairy Sci.* 87:1815-1819.
- Bimie, J. W., R. E. Agnew, and F. J. Gordon. 2000. The influence of body condition on the fasting energy metabolism of nonpregnant, nonlactating dairy cows. *J. Dairy Sci.* 83:1217-1223.
- Boerman, J. P., S. B. Potts, M. J. VandeHaar, M. S. Allen, and A. L. Lock. 2015. Milk production responses to a change in dietary starch concentration vary by production level in dairy cattle. *J. Dairy Sci.* 98:4698-4706.
- Brosh, A., Z. Henkin, E. D. Ungar, A. Dolev, A. Orlov, Y. Yehuda, and Y. Aharoni. 2006. Energy cost of cows' grazing activity: Use of the heart rate method and the Global Positioning System for direct field estimation. *J. Anim. Sci.* 84:1951-1967.
- Brosh, A., Z. Henkin, E. D. Ungar, A. Dolev, A. Shabtay, A. Orlov, Y. Yehuda, and Y. Aharoni. 2010. Energy cost of activities and locomotion of grazing cows: A repeated study in larger plots. *J. Anim. Sci.* 88(1):315-323.
- Brown, E. G., M. J. VandeHaar, K. M. Daniels, J. S. Liesman, L. T. Chapin, D. H. Keisler, and M. S. Weber Nielsen. 2005. Effect of increasing energy and protein intake on body growth and carcass composition of heifer calves. *J. Dairy Sci.* 88:585-594.
- Cammell, S. B., D. E. Beever, J. D. Sutton, J. France, G. Alderman, and D. J. Humphries. 2000a. An examination of energy utilization in lactating dairy cows receiving a total mixed ration based on maize silage. *Anim. Sci.* 71:585-596.
- Cammell, S. B., J. D. Sutton, D. E. Beever, D. J. Humphries, and R. H. Phipps. 2000b. The effect of crop maturity on the nutritional value of maize silage for lactating dairy cows 1: Energy and nitrogen utilization. *Anim. Sci.* 71:381-390.
- Capper, J. L., and D. E. Bauman. 2013. The role of productivity in improving the environmental sustainability of ruminant production systems. *Annu. Rev. Anim. Biosci.* 1:469-489.
- Capper, J. L., R. A. Cady, and D. E. Bauman. 2009. The environmental impact of dairy production: 1944 compared with 2007. *J. Anim. Sci.* 87:2160-2167.
- Chillard, Y., M. Cisse, R. LeFaivre, and B. Remond. 1991. Body composition of dairy cows according to lactation stage, somatotropin treatment, and concentrate supplementation. *J. Dairy Sci.* 74:3103-3116.
- Connor, E. E. 2015. Invited review: Improving feed efficiency in dairy production: Challenges and possibilities. *Animal.* 9:395-408.
- Crocker, L. M., E. J. DePeters, J. G. Fadel, S. E. Essex, H. Perez-Monti, and S. J. Taylor. 1998. Ash content of detergent fibers in feeds, digesta, and feces and its relevance in fiber digestibility calculations. *J. Dairy Sci.* 81:1010-1014.
- Dado, R. G., D. R. Mertens, and G. E. Shook. 1993. Metabolizable energy and absorbed protein requirements for milk component production. *J. Dairy Sci.* 76:1575-1588.
- Davis Rincker, L. E., M. S. Weber Nielsen, L. T. Chapin, J. S. Liesman, and M. J. Vandehaar. 2008. Effects of feeding prepubertal heifers a high-energy diet for three, six, or twelve weeks on feed intake, body growth and fat deposition. *J. Dairy Sci.* 91:1913-1925.

- de Souza, R., R. J. Tempelman, M. S. Allen, W. Weiss, J. Bernard, and M. J. VandeHaar. 2018. Predicting nutrient digestibility in high-producing lactating dairy cow. *J. Dairy Sci.* 101:1123-1135.
- de Souza, R. A., R. J. Tempelman, M. S. Allen, and M. J. VandeHaar. 2019. Updating predictions of dry matter intake of lactating dairy cows. *J. Dairy Sci.* 102:7948-7960.
- DePeters, E. J., and J. D. Ferguson. 1992. Nonprotein nitrogen and protein distribution in the milk of cows. *J. Dairy Sci.* 75:3192-3209.
- Dhakal, K., C. Maltecca, J. P. Cassady, G. Baloch, C. M. Williams, and S. P. Washburn. 2013. Calf birth weight, gestation length, calving ease, and neonatal calf mortality in Holstein, Jersey, and crossbred cows in a pasture system. *J. Dairy Sci.* 96:690-698.
- Dong, L. F., C. P. Ferris, D. A. McDowell, and T. Yan. 2015. Effects of diet forage proportion on maintenance energy requirement and the efficiency of metabolizable energy use for lactation by lactating dairy cows. *J. Dairy Sci.* 98(12):8846-8855.
- Edmonson, A. O., and F. G. Owen. 1989. Relation of intake to digestibility of diets containing soyhulls and distillers dried grains. *J. Dairy Sci.* 72:1786-1792.
- Edmonson, A. J., I. J. Lean, L. D. Weaver, T. Farver, and G. Webster. 1989. A body condition scoring chart for Holstein dairy cows. *J. Dairy Sci.* 72:68-78.
- Ellis, J. L., F. Qiao, and J. P. Cant. 2006. Evaluation of net energy expenditures for dairy cows according to body weight changes over a full lactation. *J. Dairy Sci.* 89:1546-1557.
- Ferraretto, L. F., P. M. Crump, and R. D. Shaver. 2013. Effect of cereal grain type and com grain harvesting and processing methods on intake, digestion, and milk production by dairy cows through a meta-analysis. *J. Dairy Sci.* 96(1):533-550.
- Ferrell, C. L. 1991. Maternal and fetal influences on uterine and conceptus development in the cow: 1. Growth of tissues of the gravid uterus. *J. Anim. Sci.* 69(5):1945-1953.
- Ferrell, C. L., W. N. Garrett, N. Hinman, and G. Grichting. 1976. Energy utilization by pregnant and nonpregnant heifers. *J. Anim. Sci.* 42:937-950.
- Firkins, J. L., M. L. Eastridge, N. R. St.-Pierre, and S. M. Noffsger. 2001. Effects of grain variability and processing on starch utilization by lactating dairy cows. *J. Anim. Sci.* 79(E. Suppl.):E218-E238.
- Flatt, W. P., C. E. Coppock, and L. A. Moore. 1965. Energy balance studies with dry, non-pregnant dairy cows consuming pelleted forages. *Proc. 3rd Symp. Energy Metab. Farm Anim. EAAP Publ.* 11:131.
- Foth, A. J., T. Brown-Brandl, K. J. Hanford, P. S. Miller, G. Garcia Gomes, and P. J. Kononoff. 2015. Energy content of reduced-fat dried distillers grains with solubles for lactating dairy cows. *J. Dairy Sci.* 98:7142-7152.
- Gamsworthy, P. C. 2006. Body condition score in dairy cows: Targets for production and fertility. Pp. 61-86 in *Recent Advances in Animal Nutrition*, P. C. Gamsworthy and J. Wiseman, eds. Nottingham, UK: Nottingham University Press.
- Gibb, M. J., W. E. Irving, M. S. Dhanoa, and J. D. Sutton. 1992. Changes in body components of autumn-calving Holstein Friesian cows over the first 29 weeks of lactation. *Anim. Prod.* 55:339-360.
- Gutierrez, G. G., L. M. Schake, and F. M. Byers. 1982. Whole plant grain sorghum silage processing and lasalocid effects on Stocker calf performance and rumen fermentation. *J. Anim. Sci.* 54(4):863-868.
- Hand, M. S., C. D. Thatcher, R. L. Remillard, and P. Roudebush. 2000. *Small Animal Clinical Nutrition*. 4th ed. Marceline, MO: Walsworth.
- Hanigan, M. D., C. C. Palliser, and P. Gregorini. 2009. Altering the representation of hormones and adding consideration of gestational metabolism in a metabolic cow model reduced prediction errors. *J. Dairy Sci.* 92(10):5043-5056.
- Hart, S. P. 1987. Associative effects of sorghum silage and sorghum grain diets. *J. Anim. Sci.* 64:1779-1789.
- Harvatine, K. J., J. W. Perfield, and D. E. Bauman. 2009. Expression of enzymes and key regulators of lipid synthesis is upregulated in adipose tissue during CLA-induced milk fat depression in dairy cows. *J. Nutr.* 139:849-854.
- House, W. A., and A. W. Bell. 1993. Mineral accretion in the fetus and adnexa during late gestation in Holstein cows. *J. Dairy Sci.* 76(10):2999-3010.
- Huhtanen, P., M. Rinne, and J. Nousiainen. 2009. A meta-analysis of feed digestion in dairy cows: 2. The effects of feeding level and diet composition on digestibility. *J. Dairy Sci.* 92:5031-5042.
- Ipharraguerre, I. R., R. R. Ipharraguerre, and J. H. Clark. 2002. Performance of lactating dairy cows fed varying amounts of soyhulls as a replacement for com grain. *J. Dairy Sci.* 85:2905-2912.
- Jenkins, T. C., and K. J. Harvatine. 2014. Lipid feeding and milk fat depression. *Vet. Clin. North Am. Food Anim. Pract.* 30(3):623-642.
- Kamal, M. M., M. Van Eetvelde, E. Depreester, M. Hostens, L. Vandaele, and G. Opsomer. 2014. Age at calving in heifers and level of milk production during gestation in cows are associated with the birth size of Holstein calves. *J. Dairy Sci.* 97:5448-5458.
- Kendall, C., C. Leonardi, P. C. Hoffman, and D. K. Combs. 2009. Intake and milk production of cows fed diets that differed in dietary neutral detergent fiber and neutral detergent fiber digestibility. *J. Dairy Sci.* 92:313-323.
- Kennedy, P. M., R. J. Christopherson, and L. P. Milligan. 1976. The effect of cold exposure of sheep on digestion, rumen turnover time, and efficiency of microbial synthesis. *Br. J. Nutr.* 36:231.
- Kirkland, R. M., and F. J. Gordon. 1999. The metabolizable energy requirement for maintenance and the efficiency of use of metabolizable energy for lactation and tissue gain in dairy cows offered a straw/concentrate ration. *Livestock Prod. Sci.* 61:23-31.
- Knowlton, K. F., V. A. Wilkerson, D. P. Casper, and D. R. Mertens. 2010. Manure nutrient excretion by Jersey and Holstein cows. *J. Dairy Sci.* 93:407-412.
- Komagiri, M., and R. A. Erdman. 1997. Factors affecting tissue mobilization in early lactation dairy cows: 1. Effect of dietary protein. *J. Dairy Sci.* 80:929-937.
- Komagiri, M., and R. A. Erdman. 1998. Factors affecting tissue mobilization in early lactation dairy cows: 2. Effect of dietary fat. *J. Dairy Sci.* 81:169-175.
- Koong, L. J., W. N. Garrett, and P. V. Rattray. 1975. A description of the dynamics of fetal growth in sheep. *J. Anim. Sci.* 41(4):1065-1068.
- Kurihara, M. 1996. Energy requirements and feed of dairy cows under high temperature conditions. *Jpn. Agric. Res. Q.* 30:107-112.
- Lamp, O., M. Demo, W. Otten, M. Mielenz, G. Nümborg, and B. Kuhla. 2015. Metabolic heat stress adaptation in transition cows: Differences in macronutrient oxidation between late-gestating and early-lactating German Holstein dairy cows. *PLoS ONE* 10(5):e0125264.
- Legault, C. R., and R. W. Touchberry. 1962. Heritability of birth weight and its relationship with production in dairy cattle. *J. Dairy Sci.* 45:1226-1233.
- Lopes, F., K. Ruh, and D. K. Combs. 2015. Validation of an approach to predict total-tract fiber digestibility using a standardized in vitro technique for different diets fed to high-producing dairy cows. *J. Dairy Sci.* 98(4):2596-2602.
- M'hamedi, N., S. Frouja, M. Bouallegue, R. Aloulou, S. K. Brar, and M. B. Hamouda. 2012. Dairy cattle welfare status measured by animal-linked parameters under Tunisian rearing conditions. In *Milk Production-An Up-to-Date Overview of Animal Nutrition, Management and Health*. IntechOpen.
- Mahjoubi, E., H. Amanlou, D. Zahmatkesh, M. Ghelich Khan, and N. Aghaziarati. 2009. Use of beet pulp as a replacement for barley grain to manage body condition score in over-conditioned late lactation cows. *Anim. Feed Sci. Tech.* 153:60-67.
- Meyer, M. J. 2005. Developmental, nutritional, and hormonal regulation of mammary growth, steroid receptor gene expression and chemical composition of retained tissues in prepubertal bovine. PhD diss.. Department of Animal Science, Cornell University, Ithaca, NY.

- Moe, P. W., H. F. Tyrrell, and W. P. Flatt. 1970. Partial efficiency of energy use for maintenance, lactation, body gain and gestation in the dairy cows. Pp. 65-68 in *Energy Metabolism of Farm Animals*, A. Schurch and C. Wenk, eds. Vitznau: European Association for Animal Production, Publication No. 13.
- Moe, P. W., H. F. Tyrrell, and W. P. Flatt. 1971. Energetics of body tissue mobilization. *J. Dairy Sci.* 54:548.
- Moraes, L. E., E. Kebreab, A. B. Strathe, J. Dijkstra, J. France, D. P. Casper, and J. G. Fadel. 2015. Multivariate and univariate analysis of energy balance data from lactating dairy cows. *J. Dairy Sci.* 98:4012-4029.
- Morris, D. L., J. L. Firkins, C. Lee, W. P. Weiss, and P. J. Kononoff. 2021. Relationship between urinary energy and urinary nitrogen or carbon excretion in lactating Jersey cows. *J. Dairy Sci.* 104:6727-6738.
- Morris, D. L., and P. J. Kononoff. 2021. Derivation of the maintenance energy requirements and efficiency of metabolizable energy utilization for dry and lactating Jersey cows. *J. Dairy Sci.* 104:9726-9734.
- NASEM (National Academies of Sciences, Engineering, and Medicine). 2016. *Nutrient Requirements of Beef Cattle*. 8th rev. ed. Washington, DC: The National Academies Press.
- NRC (National Research Council). 1978. *Nutrient Requirements of Dairy Cattle*. 5th rev. ed. Washington, DC: National Academy Press.
- NRC. 1981. *Effect of Environment on Nutrient Requirements of Domestic Animals*. Washington, DC: National Academy Press.
- NRC. 1989. *Nutrient Requirements of Dairy Cattle*. 6th rev. ed. Washington, DC: National Academy Press.
- NRC. 1996. *Nutrient Requirements of Beef Cattle*. 7th rev. ed. Washington, DC: National Academy Press.
- NRC. 2001. *Nutrient Requirements of Dairy Cattle*. 7th rev. ed. Washington, DC: National Academy Press.
- Olson, K. M., B. G. Cassell, A. J. McAllister, and S. P. Washburn. 2009. Dystocia, stillbirth, gestation length, and birth weight in Holstein, Jersey, and reciprocal crosses from a planned experiment. *J. Dairy Sci.* 92(12):6167-6175.
- Oltjen, J. W., and J. L. Beckett. 1996. Role of ruminant livestock in sustainable agricultural systems. *J. Anim. Sci.* 74:1406-1409.
- Otto, K. A., J. D. Ferguson, D. G. Fox, and C. J. Sniffen. 1991. Relationship between body condition score and composition of 9-10-11 rib tissue in Holstein dairy cows. *J. Dairy Sci.* 74:852-859.
- Potts, S. B., J. P. Boerman, A. L. Lock, M. S. Allen, and M. J. VandeHaar. 2017. Relationship between residual feed intake and digestibility for lactating Holstein cows fed high and low starch diets. *J. Dairy Sci.* 100:265-278.
- Pryce, J. E., O. Gonzalez-Recio, G. Nieuwhof, W. J. Wales, M. P. Coffey, B. J. Hayes, and M. E. Goddard. 2015. Hot topic: Definition and implementation of a breeding value for feed efficiency in dairy cows. *J. Dairy Sci.* 98:7340-7350.
- Radcliff, R. P., M. J. VandeHaar, A. L. Skidmore, L. T. Chapin, B. R. Radke, J. W. Lloyd, E. P. Staisiewski, and H. A. Tucker. 1997. Effect of diet and BST on heifer growth and mammary development. *J. Dairy Sci.* 80:1996-2003.
- Tebbe, A. W., M. J. Faulkner, and W. P. Weiss. 2017. Effect of partitioning the nonfiber carbohydrate fraction and neutral detergent fiber method on digestibility of carbohydrates by dairy cows. *J. Dairy Sci.* 100:6218-6228.
- Tempelman, R. J., D. M. Spurlock, M. Coffey, R. F. Veerkamp, L. E. Armentano, K. A. Weigel, Y. de Haas, C. R. Staples, E. E. Connor, Y. Lu, and M. J. VandeHaar. 2015. Heterogeneity in genetic and nongenetic variation and energy sink relationships for residual feed intake across research stations and countries. *J. Dairy Sci.* 98:2013-2026.

- Tyrrell, H. F., and P. W. Moe. 1972. Net energy values for lactation of a high and low concentrate ration containing com silage. *J. Dairy Sci.* 55:1106-1112.
- Tyrrell, H. F., and P. W. Moe. 1975. Effect of intake on digestive efficiency. *J. Dairy Sci.* 58:1151-1163.
- Tyrrell, H. F., and J. T. Reid. 1965. Prediction of the energy value of cow's milk. *J. Dairy Sci.* 48:1215-1223.
- Tyrrell, H. F., C. K. Reynolds, and H. D. Baxter. 1991. Utilization of dietary energy by Jersey compared to Holstein cows during the lactation cycle. *Proc. 12th Symp. Energy Metab. Farm Anim. EAAP Publ.* 58.
- Uddin, M. E., O. I. Santana, K. A. Weigel, and M. A. Wattiaux. 2020. Enteric methane, lactation performances, digestibility, and metabolism of nitrogen and energy of Holsteins and Jerseys fed 2 levels of forage fiber from alfalfa silage or com silage. *J. Dairy Sci.* 103:6087-6099.
- Van Es, A. J. H. 1961. Between animal variation in the amount of energy required for the maintenance of cows. Doctoral diss., Pudoc, Wageningen. The Netherlands.
- VandeHaar, M. J. 1998. Efficiency of nutrient use and relationship to profitability on dairy farms. *J. Dairy Sci.* 81:272-282.
- VandeHaar, M. J., L. E. Armentano, K. Weigel, D. M. Spurlock, R. J. Tempelman, and R. Veerkamp. 2016. Harnessing the genetics of the modern dairy cow to continue improvements in feed efficiency. *J. Dairy Sci.* 99:4941-4954.
- Veerkamp, R. F., A. R. Cromie, and G. Simm. 1995. Variance-components for residual feed-intake in dairy cows. *Livestock Prod. Sci.* 41(2): 111-120.
- Voelker, J. A., G. M. Burato, and M. S. Allen. 2002. Effects of pretrial milk yield on responses of feed intake, digestion, and production to dietary forage concentration. *J. Dairy Sci.* 85:2650-2661.
- Waldo, D. R., H. F. Tyrrell, A. V. Capuco, and C. E. Rexroad, Jr. 1997. Components of growth in Holstein heifers fed either alfalfa or com silage diets to produce two daily gains. *J. Dairy Sci.* 80:1674-1684.
- Waltner, S. S., J. P. McNamara, J. K. Hillers, and D. L. Brown. 1994. Validation of indirect measures of body fat in lactating cows. *J. Dairy Sci.* 77:2570-2579.
- Weiss, W. P., H. R. Conrad, and N. R. S. Pierre. 1992. A theoretically-based model for predicting total digestible nutrient values of forages and concentrates. *Anim. Feed Sci. Technol.* 39:95-110.
- West, J. W. 1994. Interactions of energy and bovine somatotropin with heat stress. *J. Dairy Sci.* 77:2091-2102.
- Wheelock, J. B., R. P. Rhoads, M. J. VanBaale, S. R. Sanders, and L. H. Baumgard. 2010. Effects of heat stress on energetic metabolism in lactating Holstein cows. *J. Dairy Sci.* 93(2):644-655.
- White, R. R., Y. Roman-Garcia, J. L. Firkins, M. J. VandeHaar, L. E. Armentano, W. P. Weiss, T. McGill, R. Garrett, and M. D. Hanigan. 2017. Evaluation of the National Research Council (2001) dairy model and derivation of new prediction equations: 1. Digestibility of fiber, fat, protein, and nonfiber carbohydrate. *J. Dairy Sci.* 100:3591-3610.
- Wildman, E. E., G. M. Jones, P. E. Wagner, R. L. Boman, H. F. Troutt, Jr., and T. N. Lesch. 1982. A dairy cow body condition scoring system and its relationship to standard production characteristics. *J. Dairy Sci.* 65:495-501.
- Xue, B., T. Yan, C. F. Ferris, and C. S. Mayne. 2011. Milk production and energy efficiency of Holstein and Jersey-Holstein crossbred dairy cows offered diets containing grass silage. *J. Dairy Sci.* 94:1455-1464.
- Young, B. W. 1976. Effects of cold environments on nutrient requirements of ruminants. Pp. 491-496 in *Proc. 1st Int. Symposium on Feed Composition, Animal Nutrient Requirements, and Computerization of Diets*, P. V. Fonnesbeck, L. F. Harris, and L. C. Kearl, eds. Logan: Utah State University.